

Přednáška 02

Odrazivost a rozptyl světla

Parametrizace a zachycení materiálů pomocí BRDF a BSSRDF

Jiří Filip

filipj@utia.cas.cz

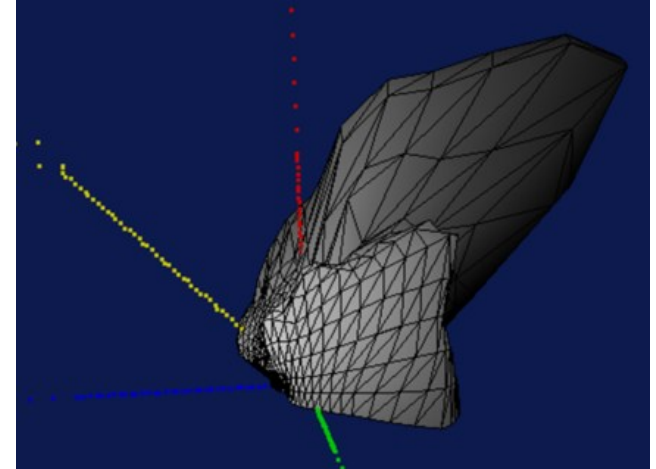


Czech Academy of Sciences



BRDF measurement

- Sampling of reflectance lobes
- Measurement process:
 - Speed vs. fidelity
 - Uniform vs. adaptive sampling
 - Importance of angular parameterization
 - Often related to DOF realization of capturing device



BRDF Anisotropic vs. Isotropic

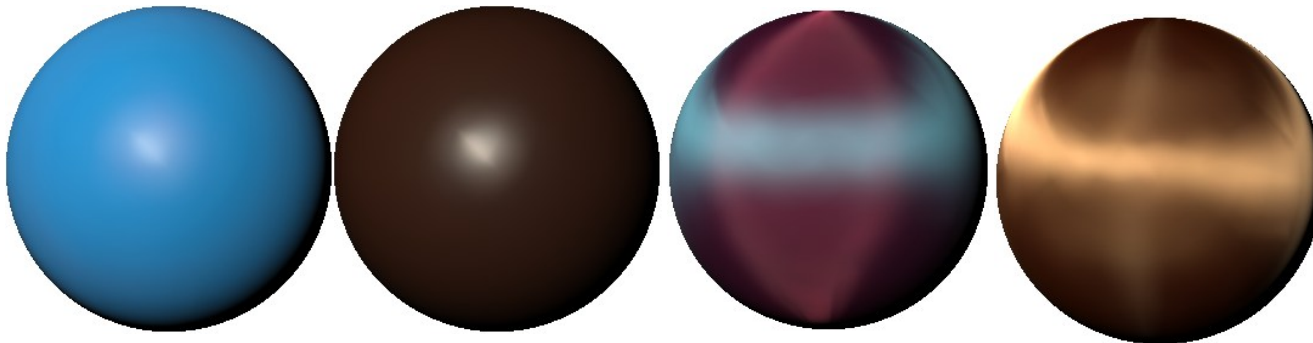
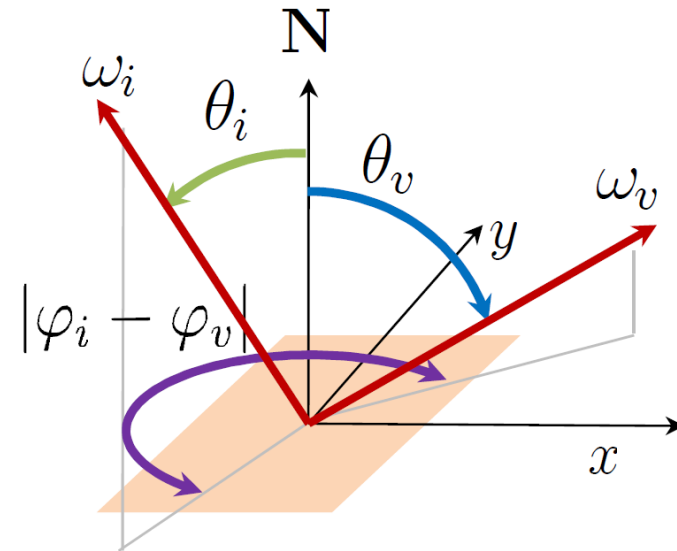
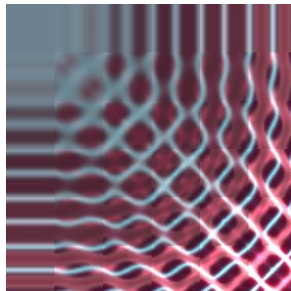
- BRDF – Bidirectional Reflectance Distribution Function
- Illumination and view dependent reflectance

$$BRDF(\lambda, \theta_i, \phi_i, \theta_v, \phi_v)$$

- Isotropic vs. anisotropic BRDF

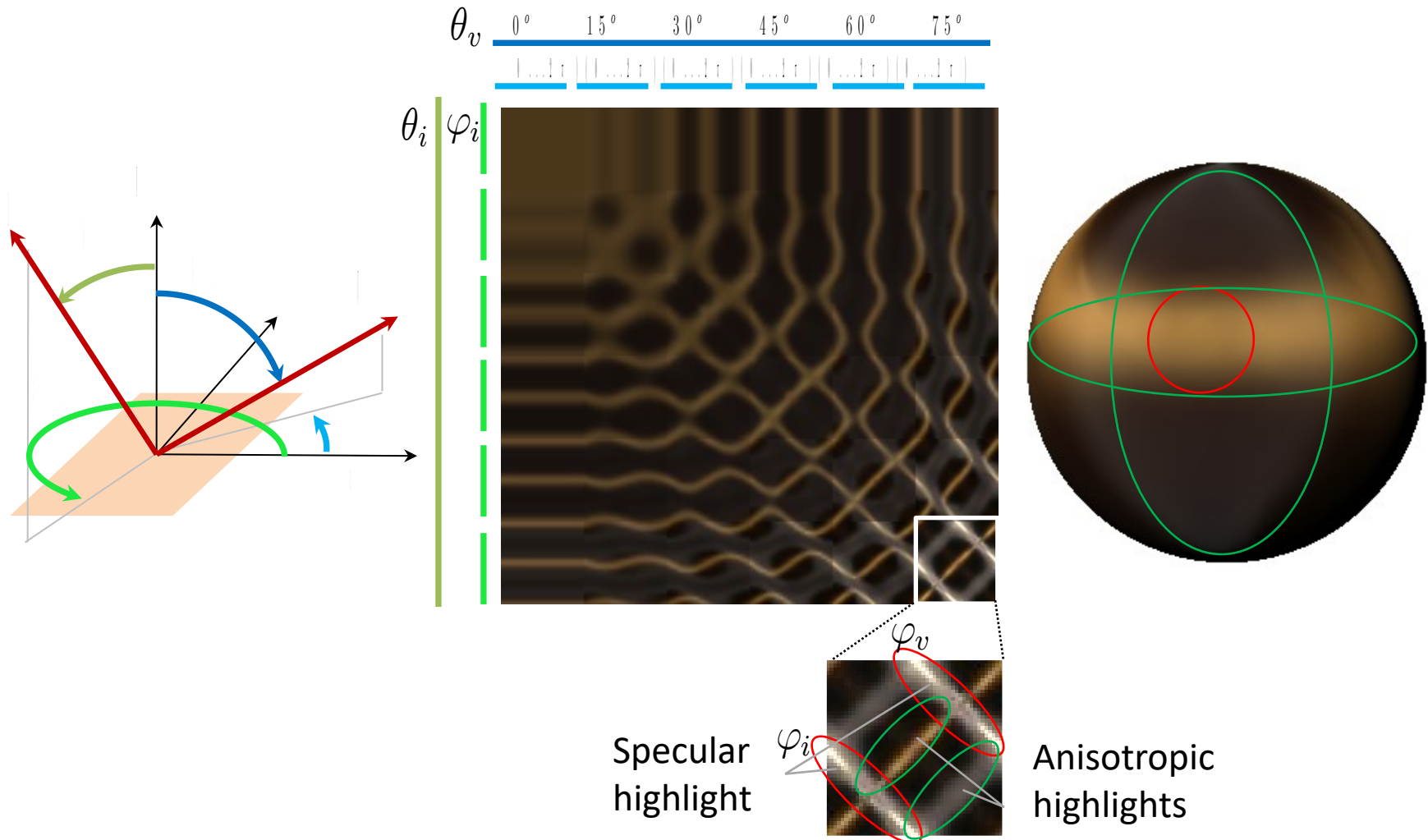
$$B(\theta_i, \theta_v, |\varphi_i - \varphi_v|)$$

$$B(\theta_i, \varphi_i, \theta_v, \varphi_v)$$

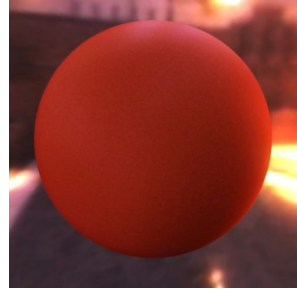


Anisotropic BRDF

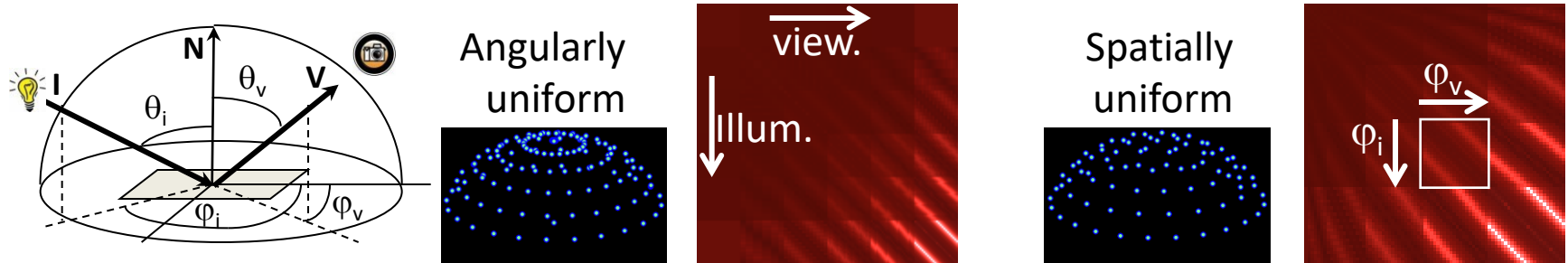
4D BRDF unfolding to 2D



Angular Parameterizations



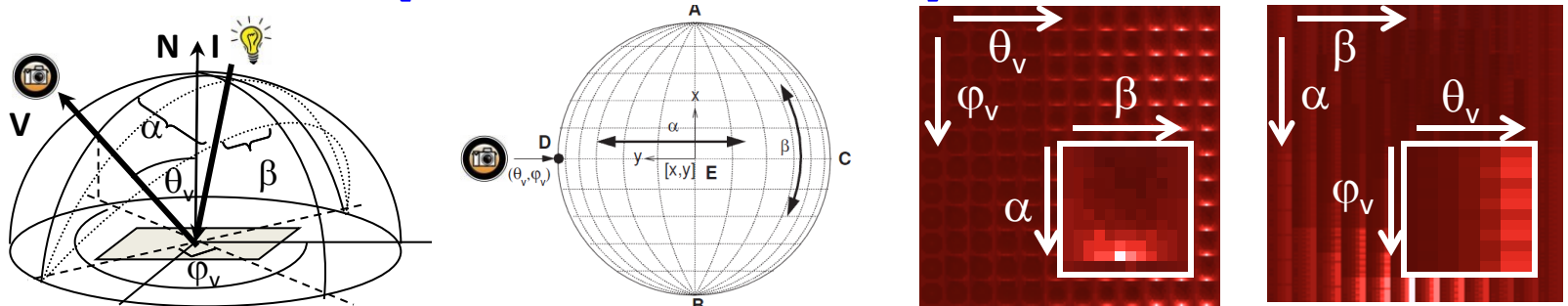
- Illumination-view



- Half-difference angles [Rusinkiewicz 98]

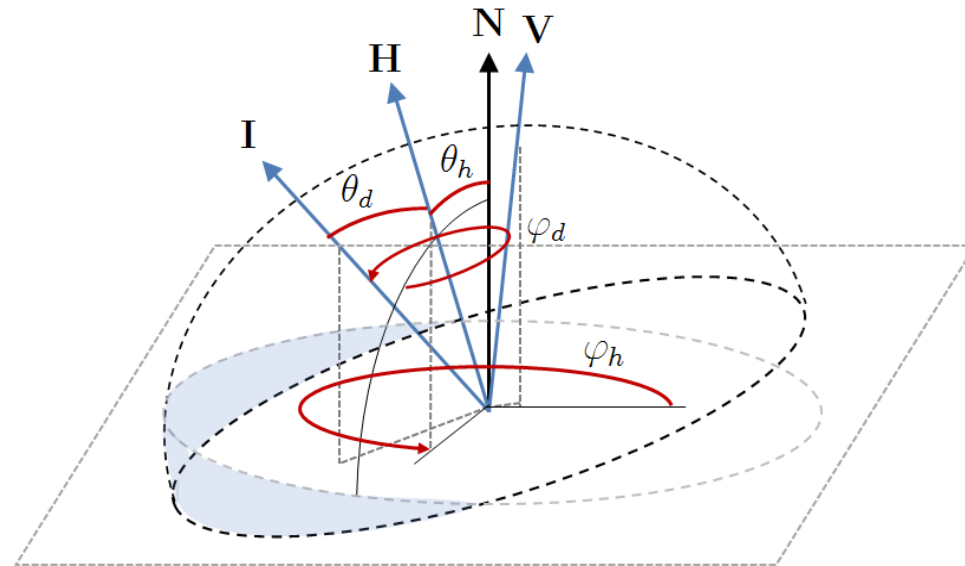


- Onion slices model [Havran et al. CGF 09]

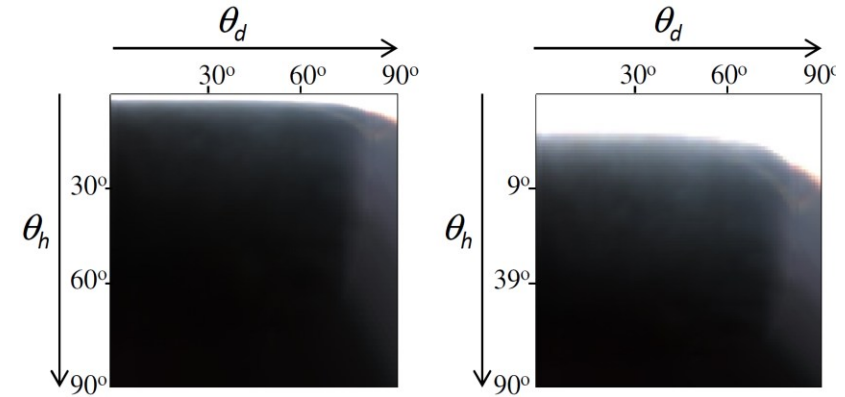


Half-Difference Parameterization

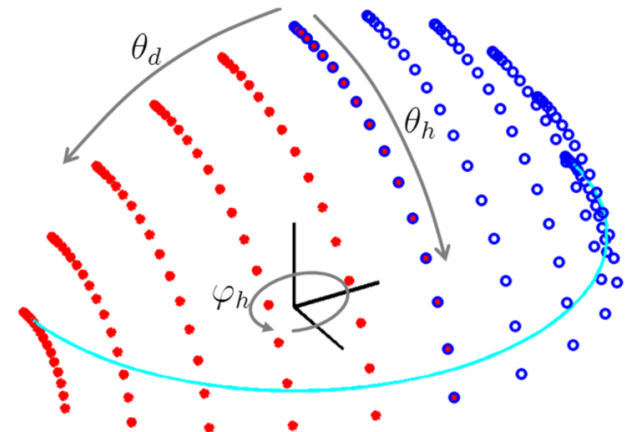
[Rusinkiewicz98]



Characteristic slice [Burley12]



Slice captured by 108 samples



specular peak

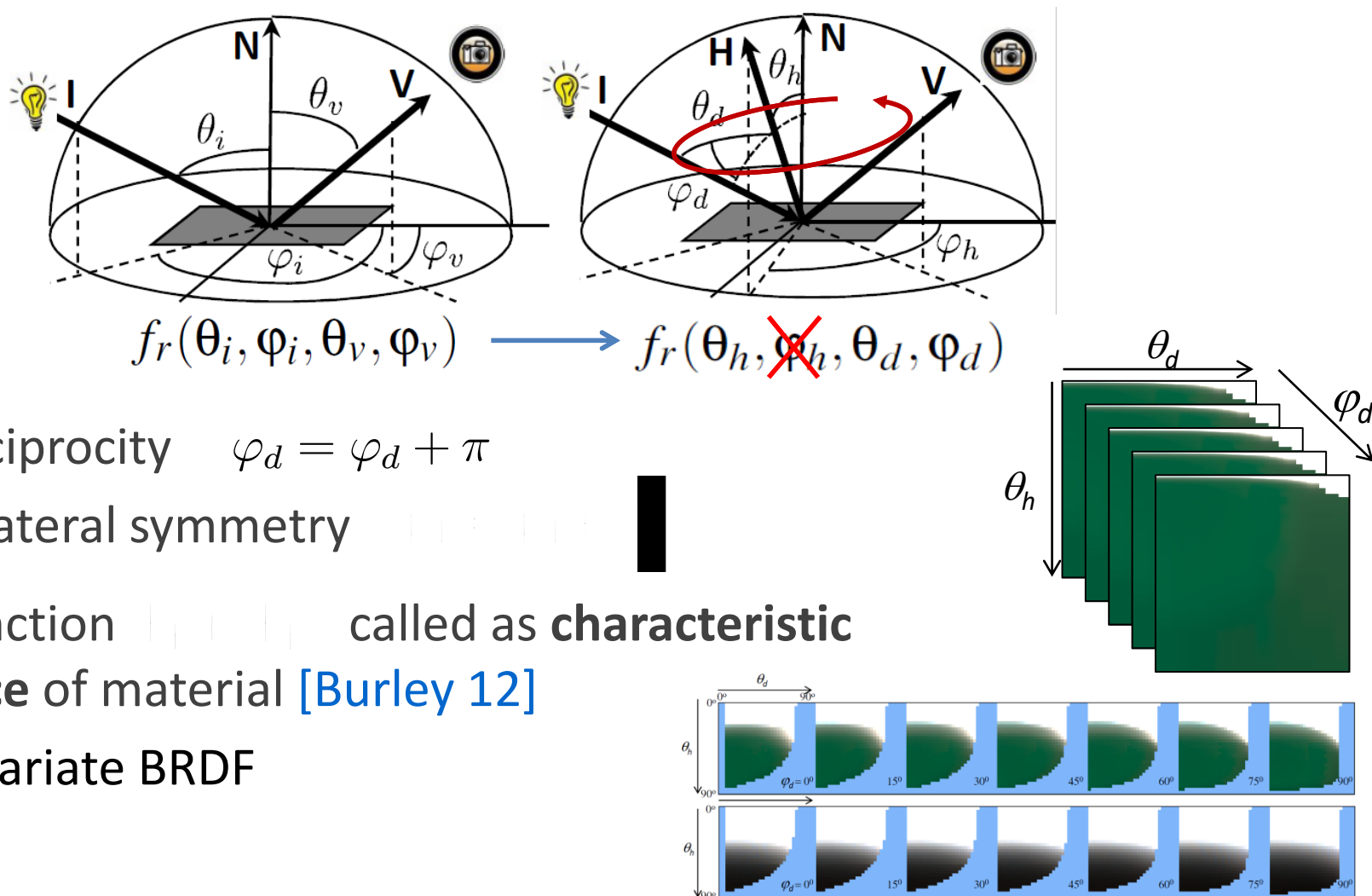
anisotropic behavior

Fresnel peak

φ_d reciprocity, symmetry

Half-Difference Parameterization

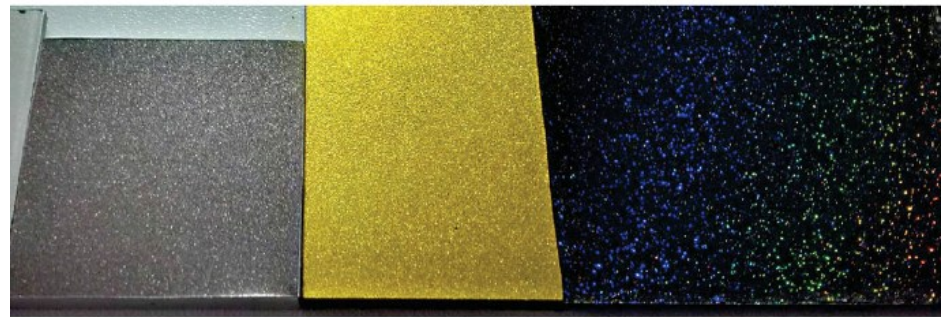
BRDF in half-difference parameterization [Rusinkiewicz 98]



- reciprocity $\varphi_d = \varphi_d + \pi$
- bilateral symmetry
- function called as **characteristic slice** of material [Burley 12]
- Bivariate BRDF

Example of Angular Parameterizations

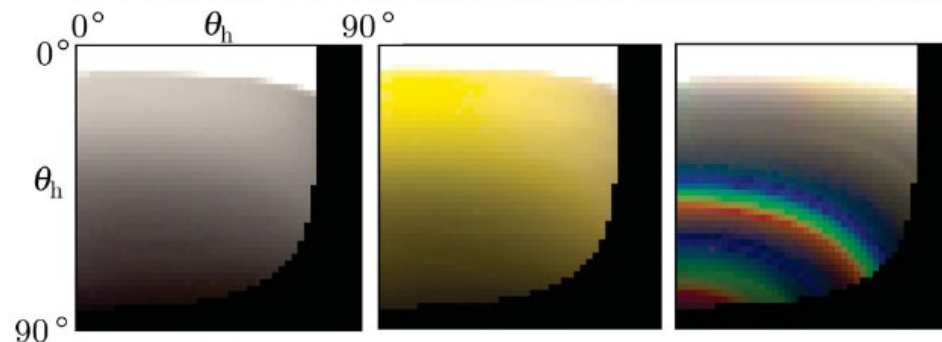
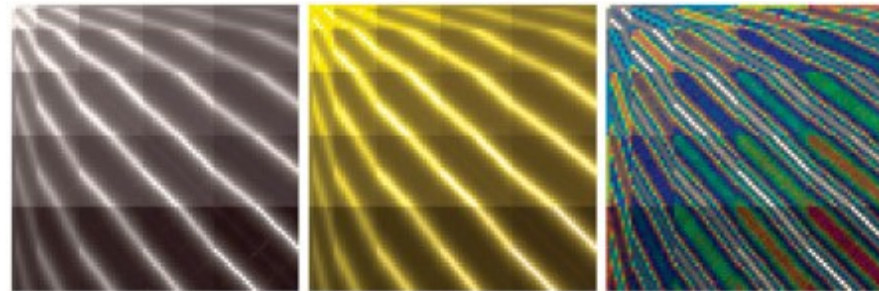
- Hemispherically uniform sampling vs. bivariate sampling of effect coatings



Silverdollar

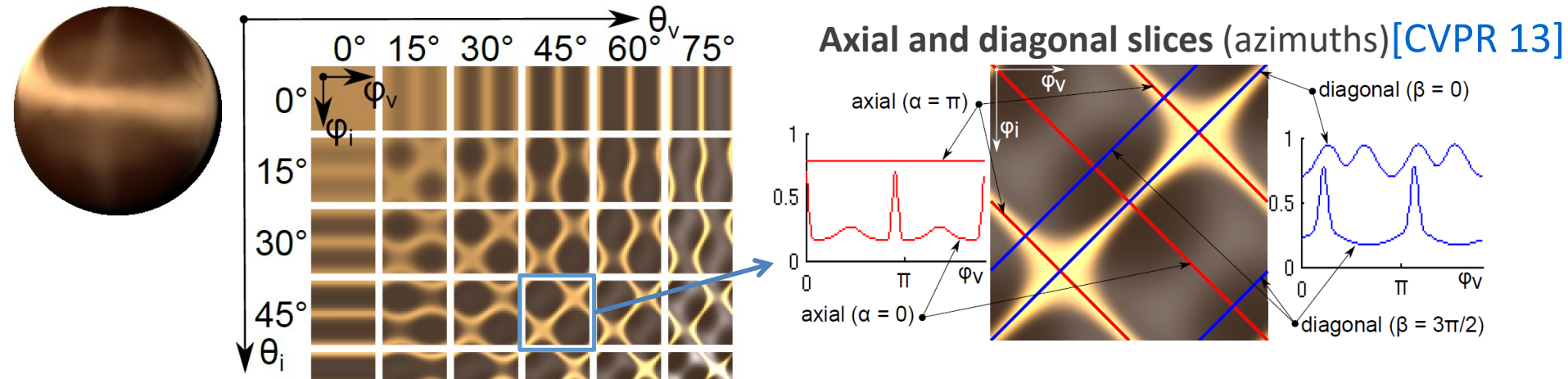
UTP

Polychromatic



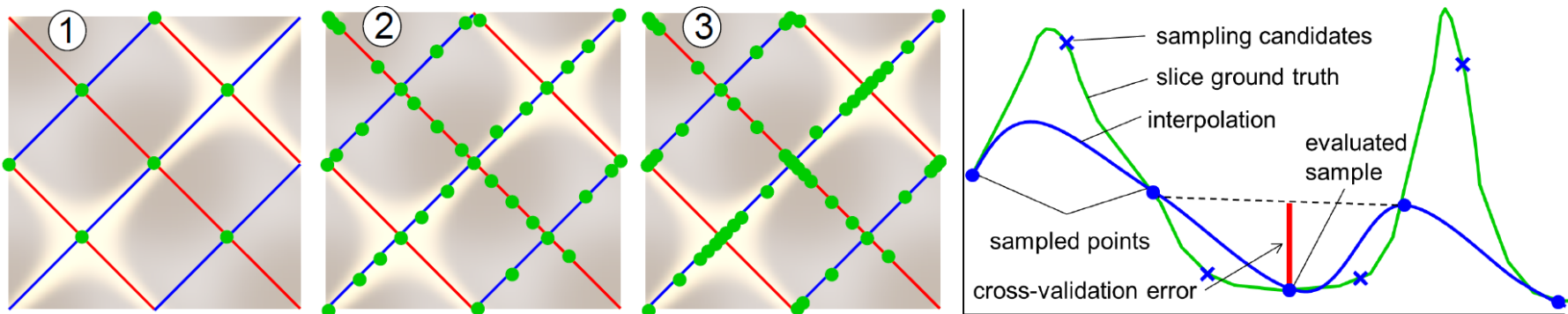
Uniform vs. adaptive sampling approaches

- BRDF slices

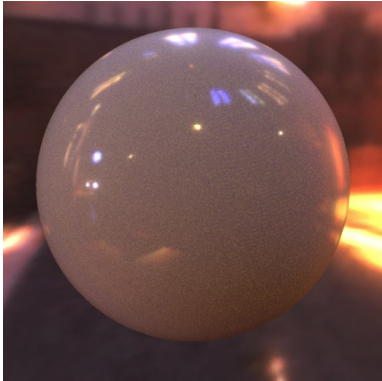


- adaptive measurement of 4D function $\Rightarrow$
adaptive measurement of 1D functions in 4D space

adaptive sampling based on a cross-validation error in control samples

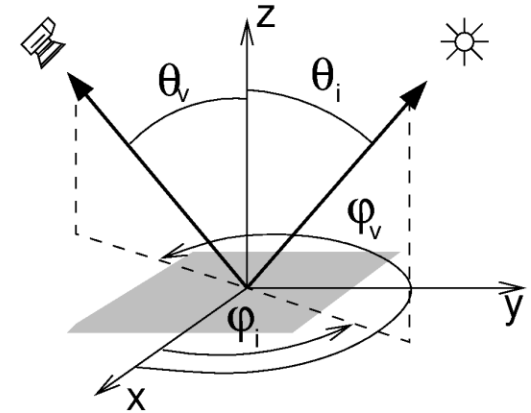


BRDF Measurement Setups Taxonomy



$$BRDF(\lambda, \theta_i, \varphi_i, \theta_v, \varphi_v)$$

5 dimensional data $\Rightarrow$ 4 dimensions depend on camera, light & sample positioning



Measurement setup with **4 mechanical degrees of freedom**:

Gonio-reflectometers

sample/light/camera **1/2/1**

Mirror-based setups

sample/light **2/2** + many views at once

Image-based setups

light/camera **1/1** + defined shape

Portable setups

compromise accuracy measurements

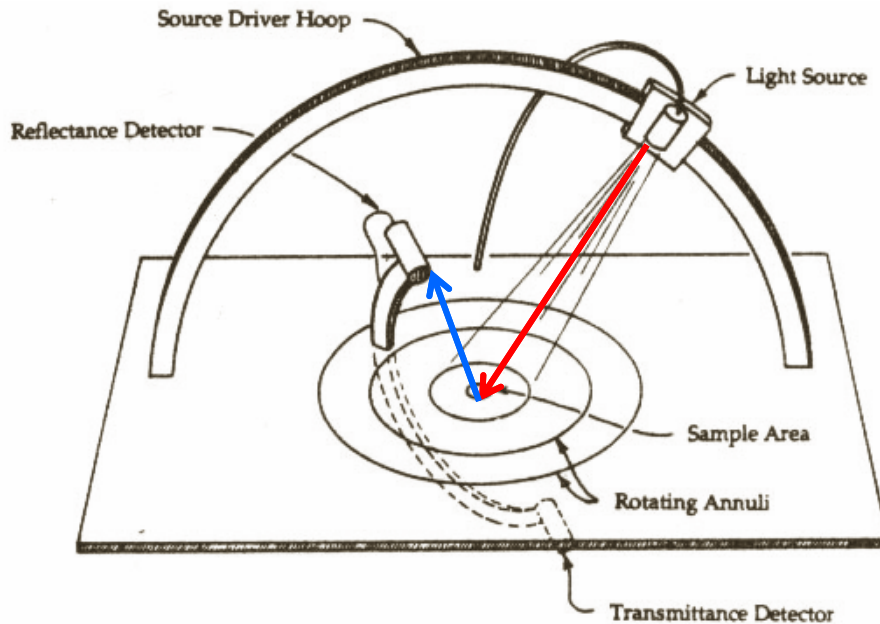
Isotropic BRDF (4 dimensional): $BRDF(\lambda, \theta_i, \theta_v, \varphi_i - \varphi_v)$

BRDF Measurement

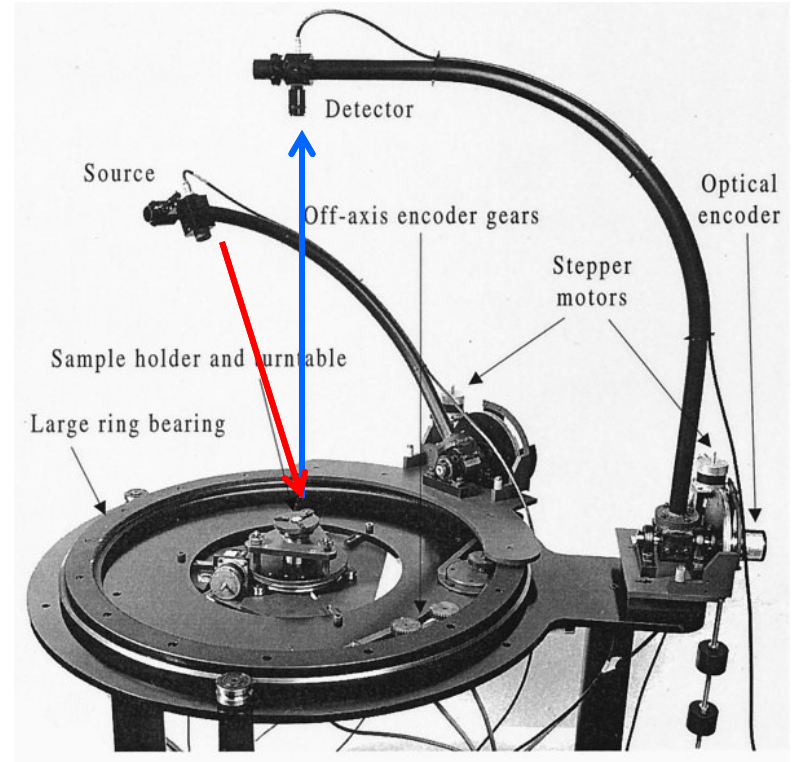
Gonio-reflectometers

Sequential sampling of 4D space $\Rightarrow$
moving sample, light & camera

[Murray & Smith JIES 90]



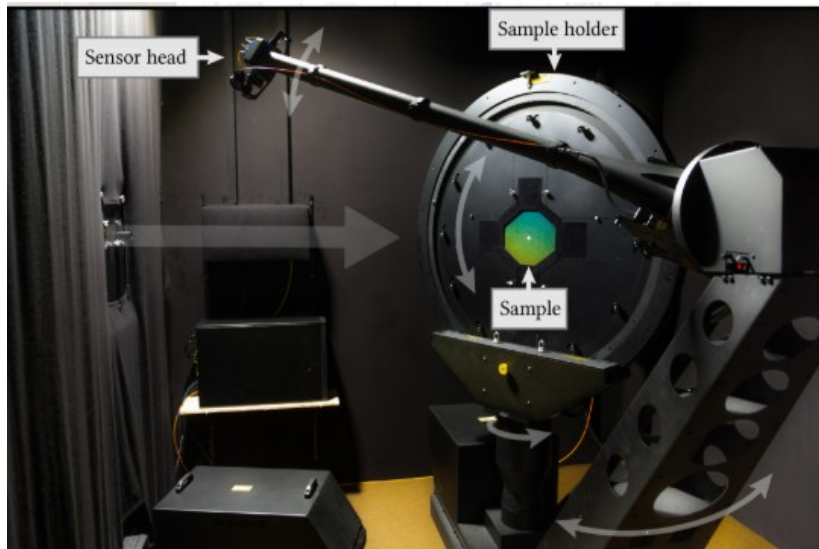
[White et al. JAO 98]



BRDF Measurement

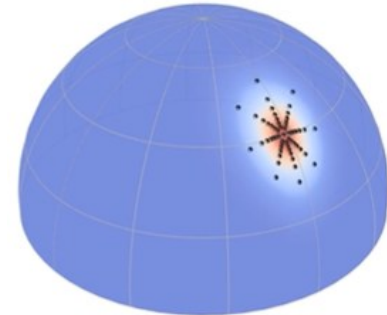
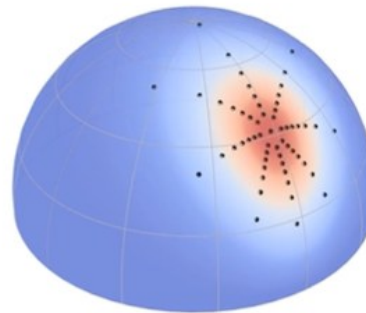
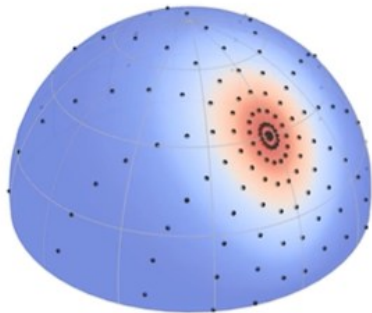
Gonio-reflectometers

[Dupuy et al. SIGA 18]



Spectral isotropic/anisotropic BRDF measurement

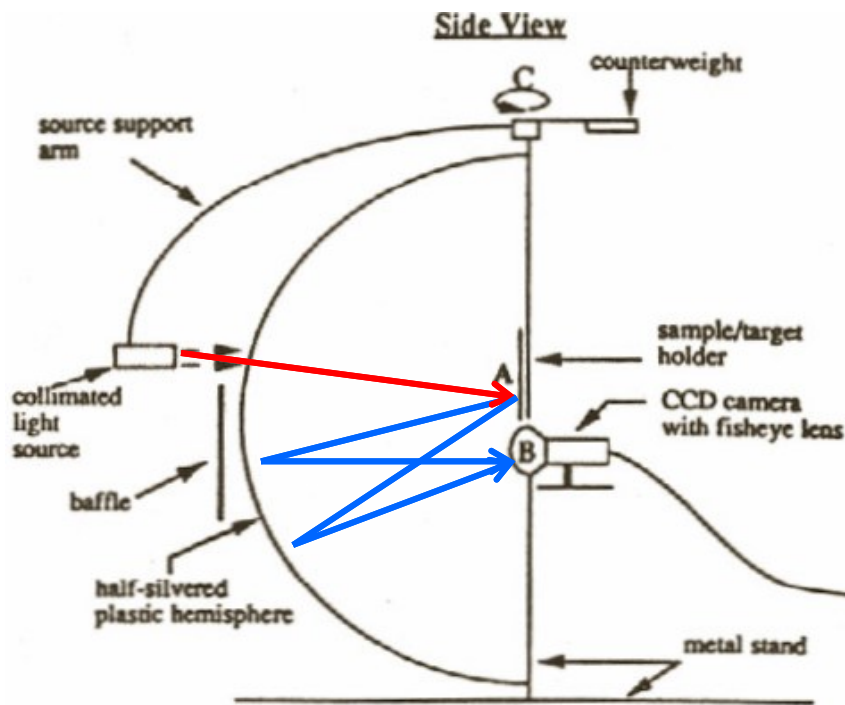
Adaptive samples parametrization



BRDF Measurement

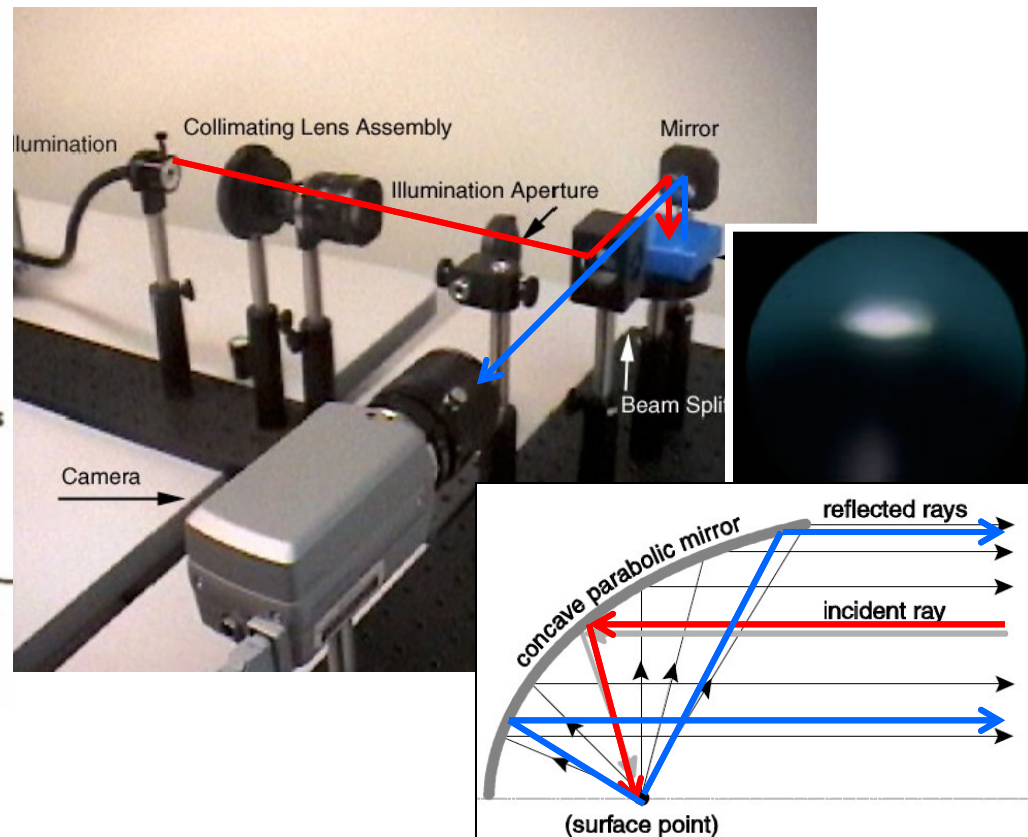
Mirror-based setups

[Ward CG 92]



Mechanical DOF reduced by multiple-views in mirror image (directional illumination)

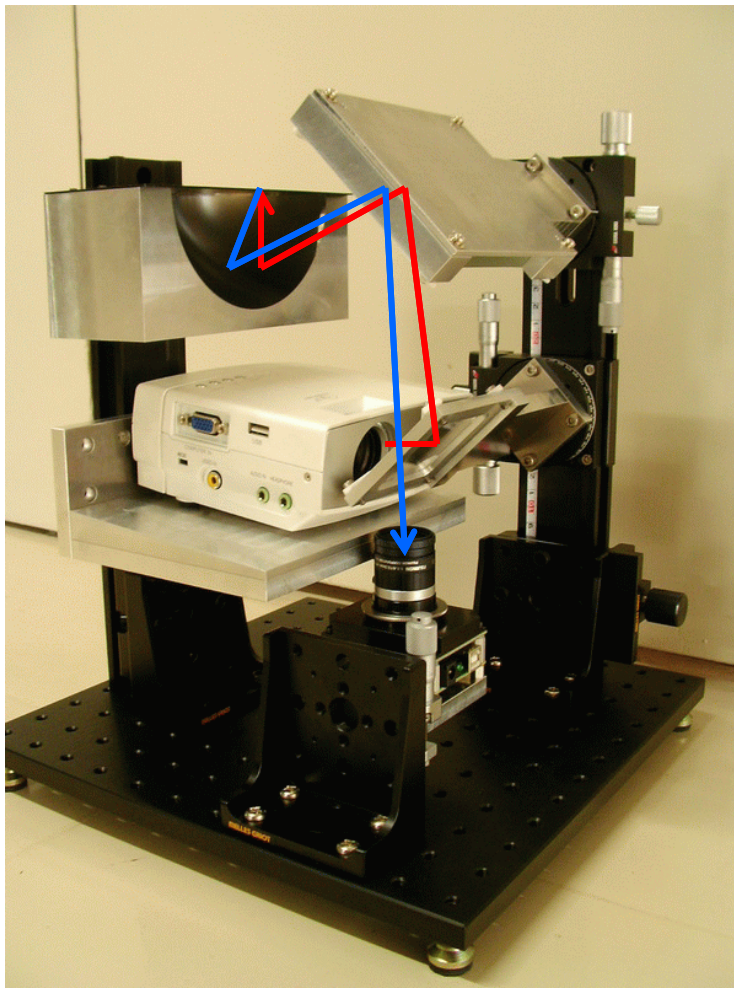
[Dana et al. ICCV 01]



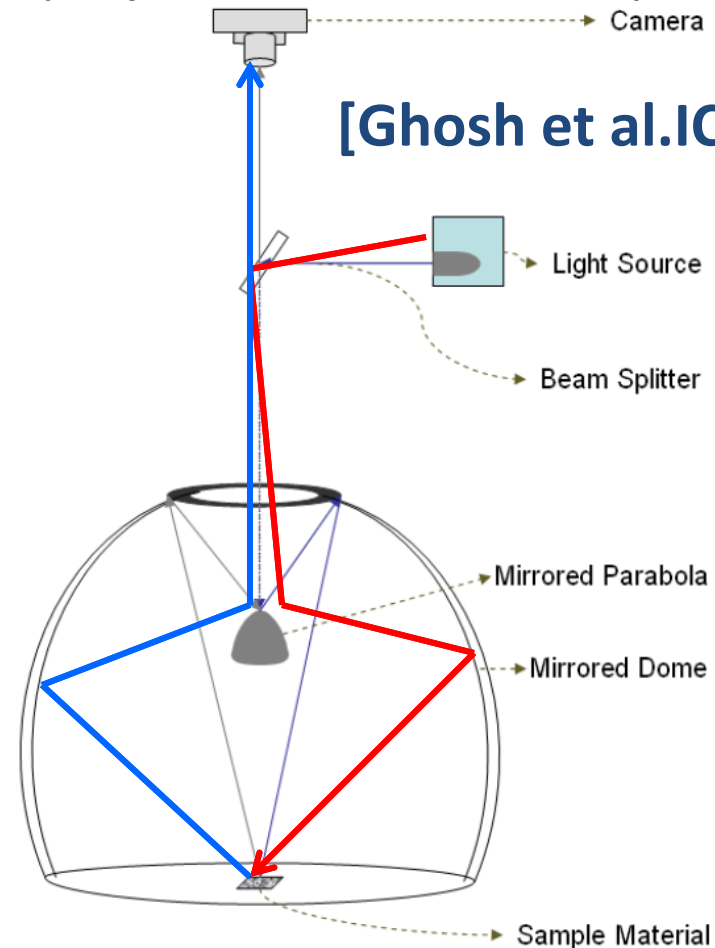
BRDF Measurement

Mirror-based setups

[Mukaigawa et al. ACCV 07]



Mechanical DOF reduced by multiple-views in mirror image (projected illumination pattern)

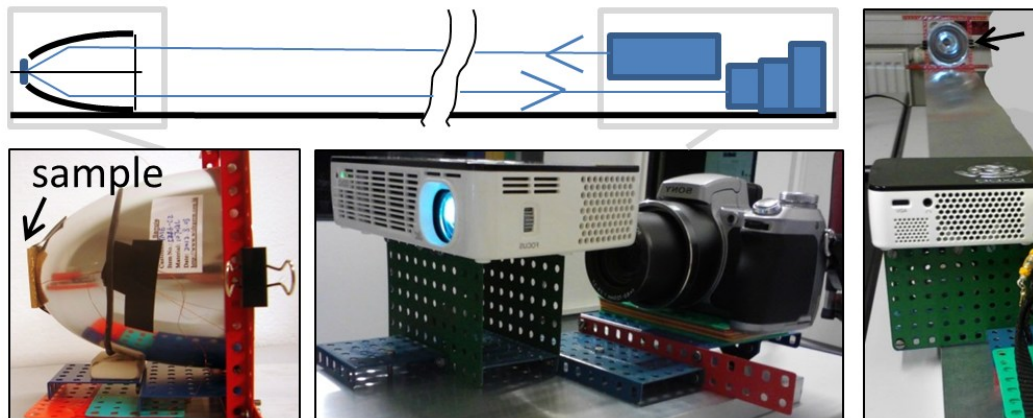


BRDF Measurement

Mirror-based setups

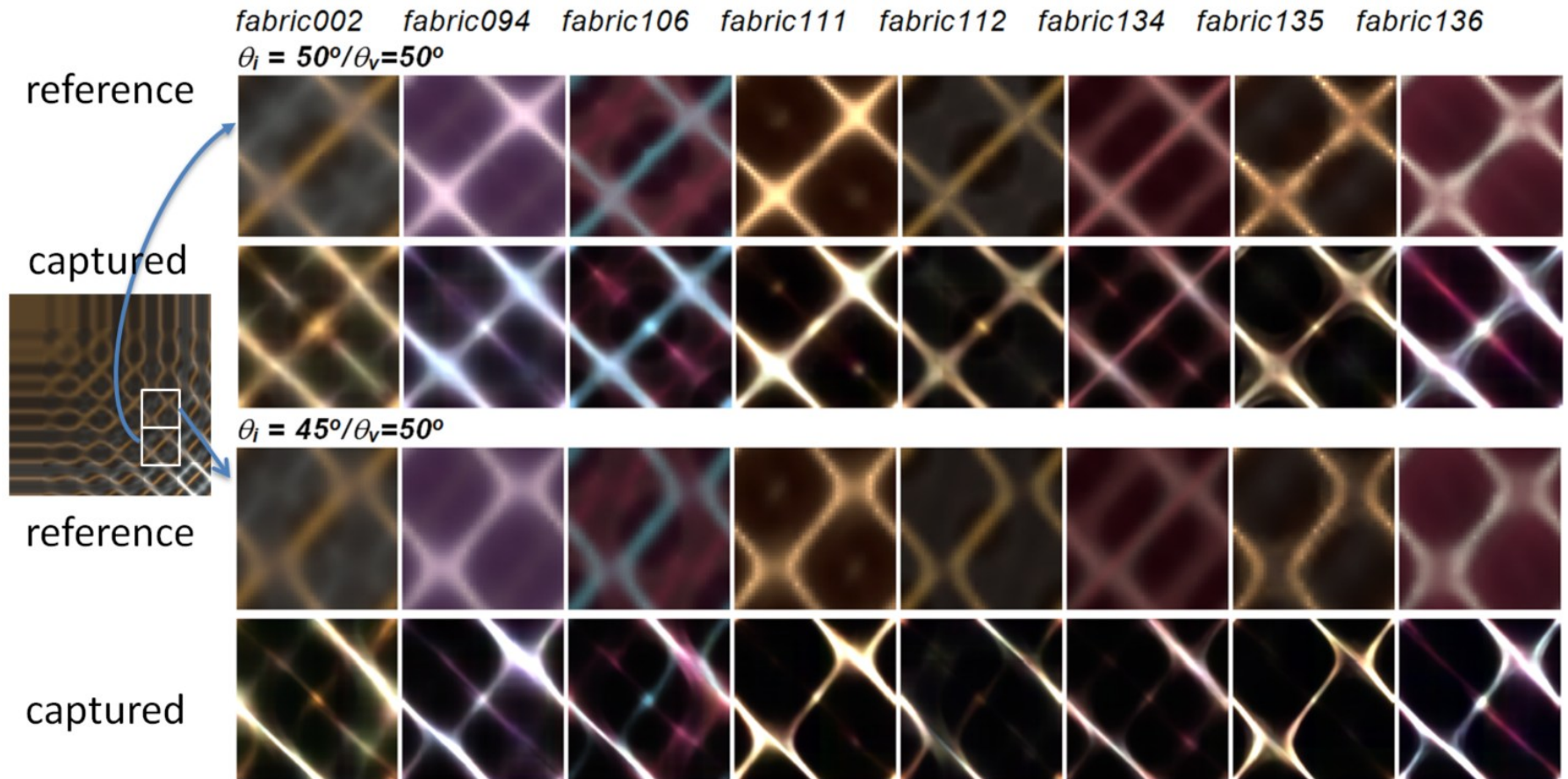
[Filip CIE 16]

- ellipsoidal reflector
- portable DSP projector AAXA P300 (300 lm)
- compact camera Sony H1



- camera – reflector distance 1.5 m
- lens close to projector emitting window $\Rightarrow$ almost identical optical paths
- light-camera-reflector alignment calibrated $\Rightarrow$ projected pattern
- measurement time 1-3 minutes

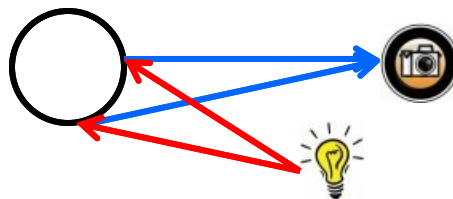
BRDF Measurement



BRDF Measurement

Image-based setups

Varying incoming
/outgoing directions
over **cylinder** image



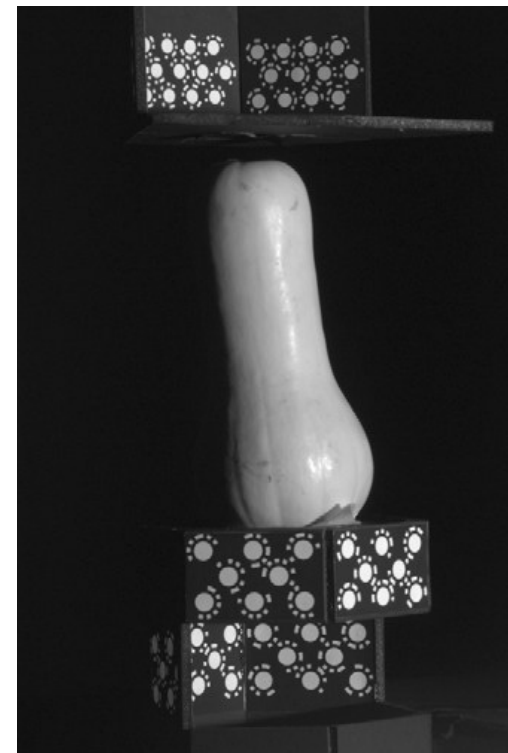
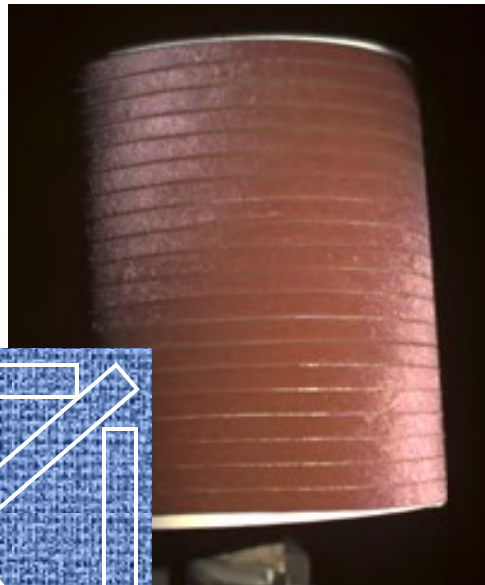
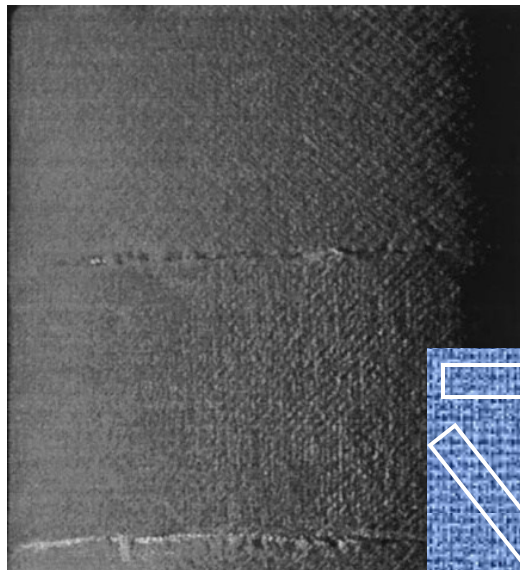
Mechanical DOF reduced by defined
sample shape (orientation)

[Marschner et al. JAO 00]

object of estimated geometry

[Lu et al. JAO 98]

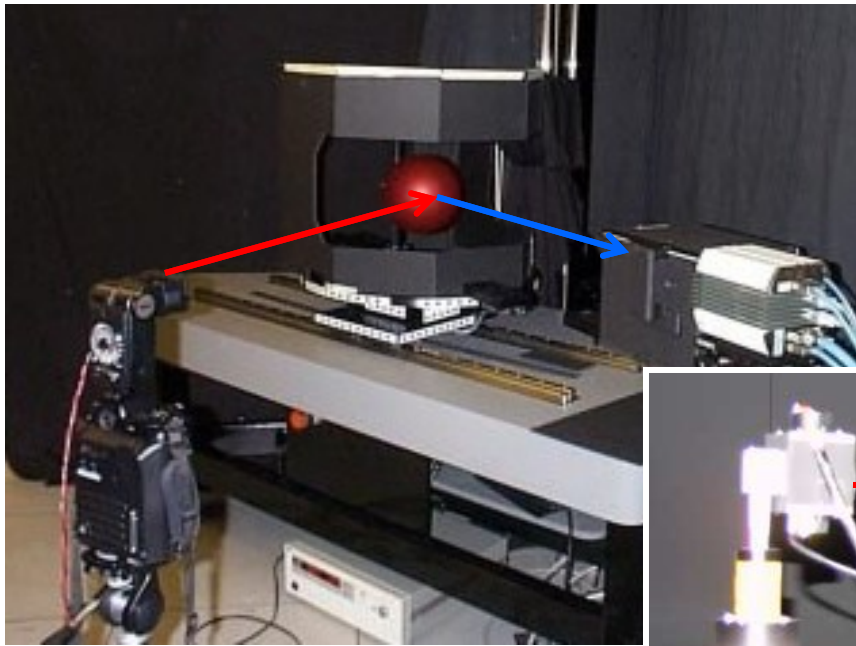
[Ngan et al. EGSR 05]



BRDF Measurement

Image-based setups

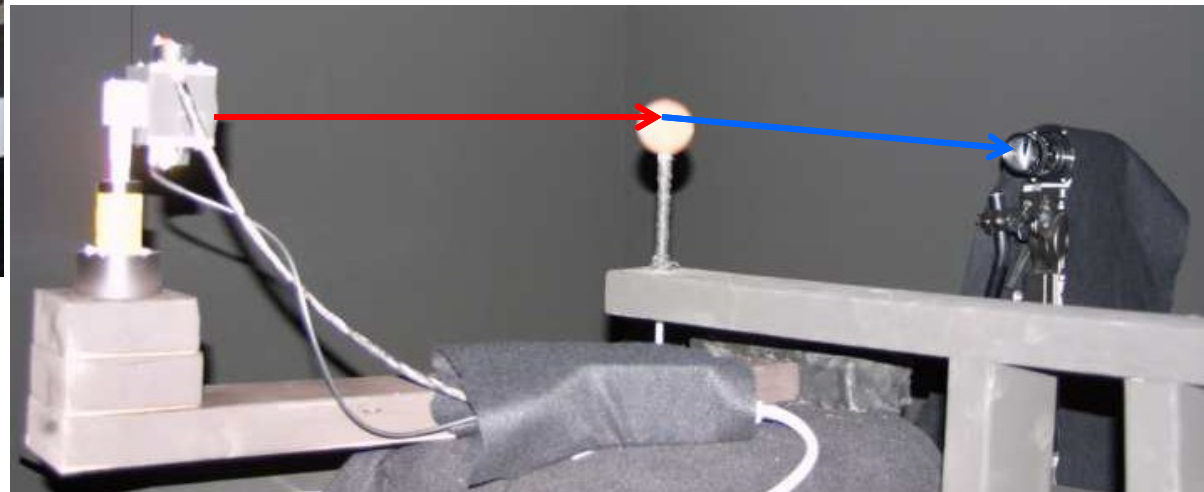
[Marschner PhD 98]



Mechanical DOF reduced by defined sample shape (orientation)

Spherical homogeneous samples

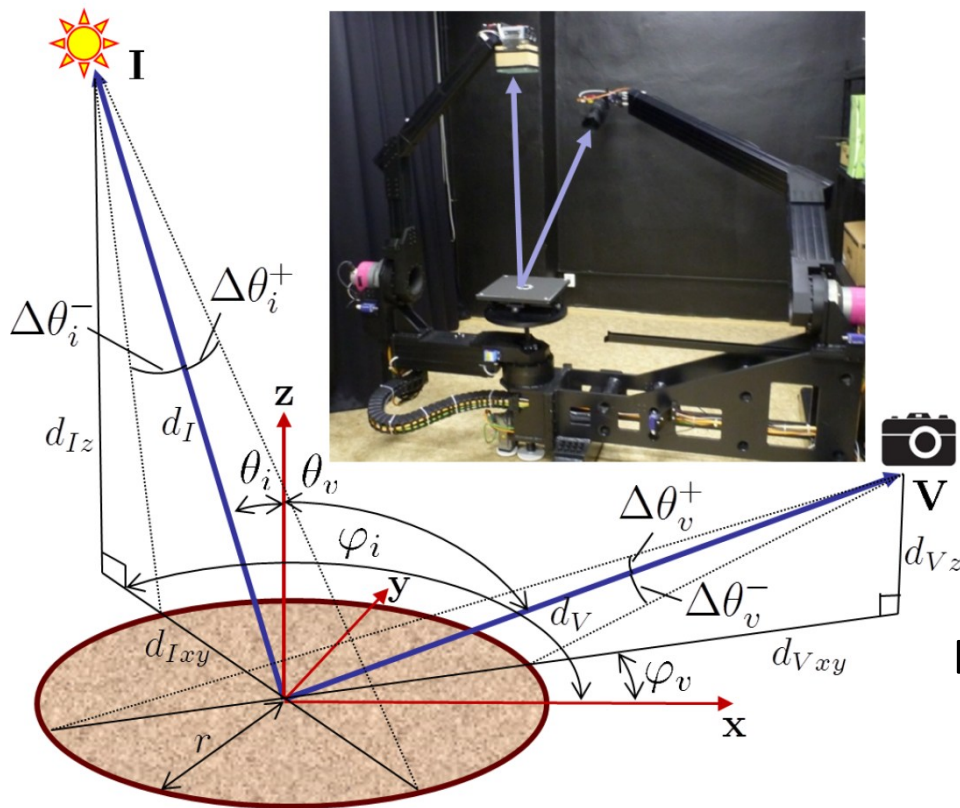
[Matusik et al. EWR 03]



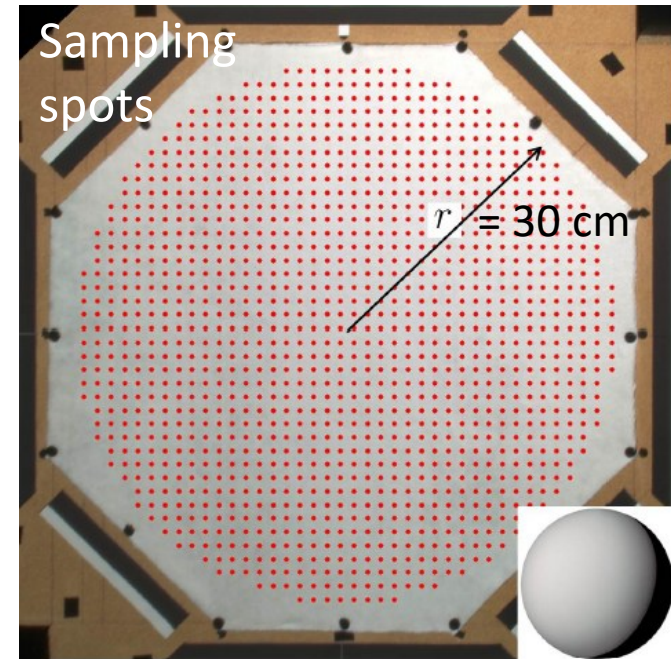
BRDF Measurement

Image-based setups

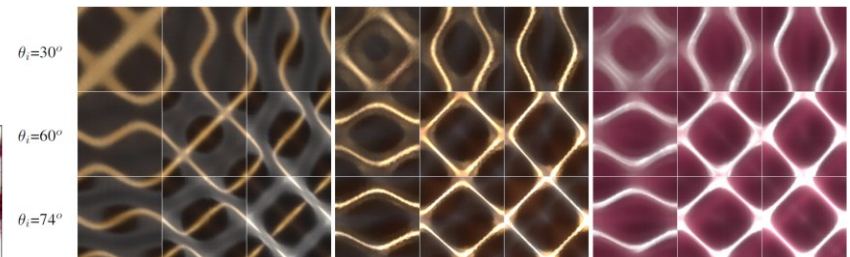
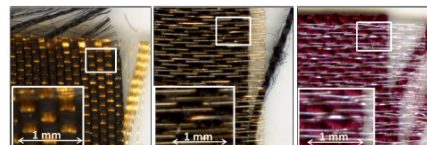
[Filip ICPR 14]



Varying angles across large material sample



BRDF with angular resolution 2 degrees



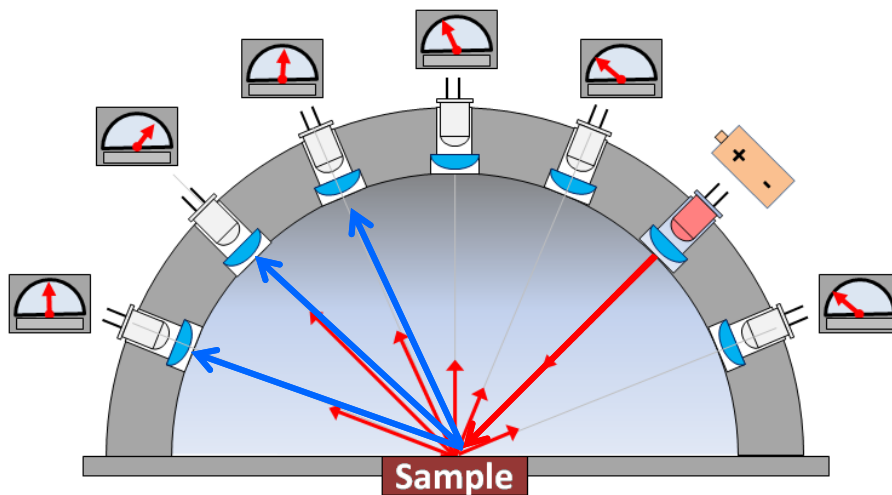
BRDF Measurement

Portable setups

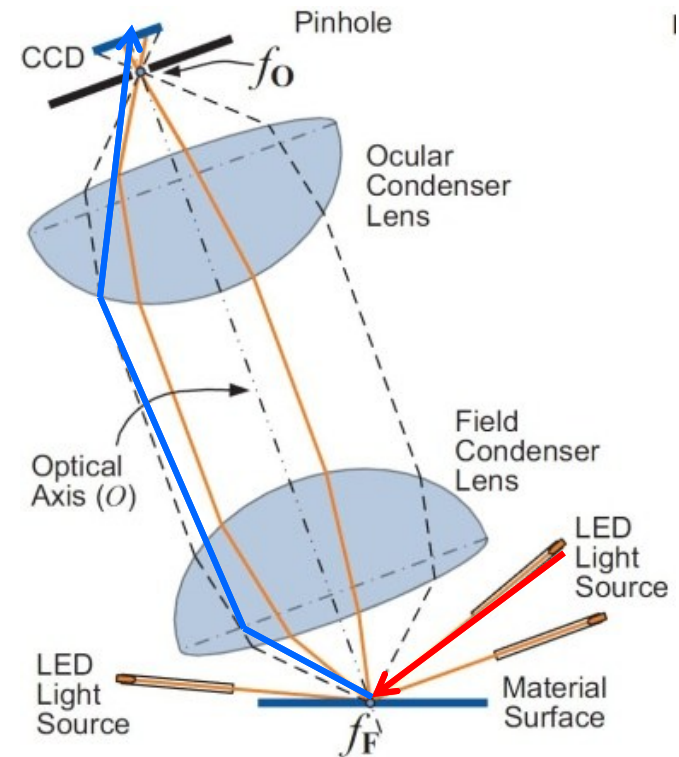
Fast measurement, compromise accuracy, limited:

- number of illumination/sensing elements
- viewing/illumination angles range

[Ben-azra et al. CVPR 08]

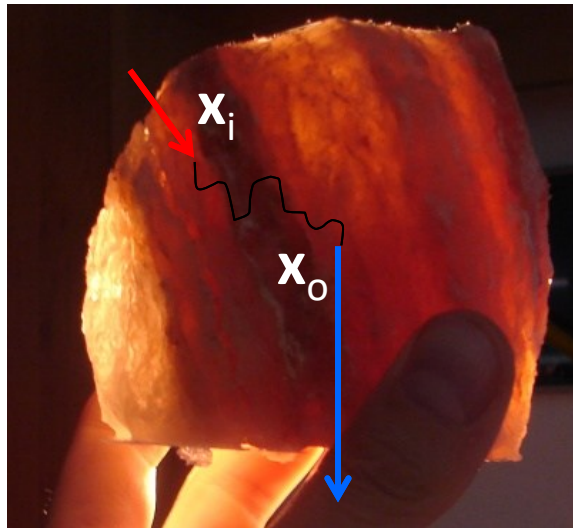


[Lan et al. CGF 10]



BSSRDF Measurement

Bidirectional subsurface-scattering reflectance distribution function



$$BSSRDF(\lambda, \underline{x_i}, \underline{y_i}, \underline{x_o}, \underline{y_o}, \underline{\theta_i}, \underline{\varphi_i}, \underline{\theta_v}, \underline{\varphi_v})$$

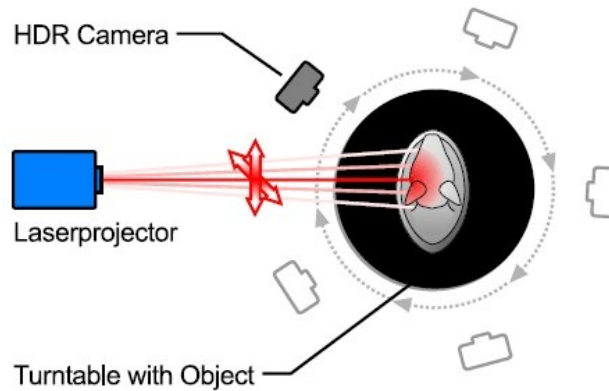
[Nicodemus et al. 77]

BTF includes scattering information,
but difficult to separate

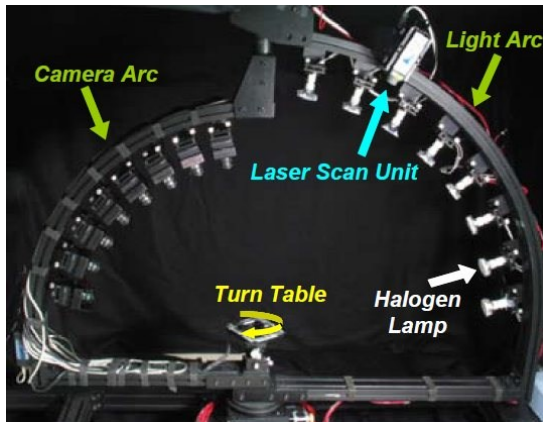
- **General 9 dimensional data** $\Rightarrow$ often called as **reflectance fields**
- Describes light transport (scattering) in material structure between any pairs of incoming and outgoing rays
- Translucent materials
- Direct measurement very sparse due to high data dimensionality

BSSRDF Measurement

[Goesele et al. TOG 04] – laser-based sparse spatially varying subsurface scattering measurement



[Tong et al. TOG 05] – BTF combined with local laser-based subsurface scattering measurement



BTF+local scat.

BTF

BTF+global scat.

BSSRDF Measurement

- Models of subsurface scattering in homogeneous dielectric materials are available, measurement of models parameters:
[Jensen et al. SIG 01] – dipole model of dielectrics, validated by scattering measurement of focused beam

Without
scattering
model



With
scattering
model

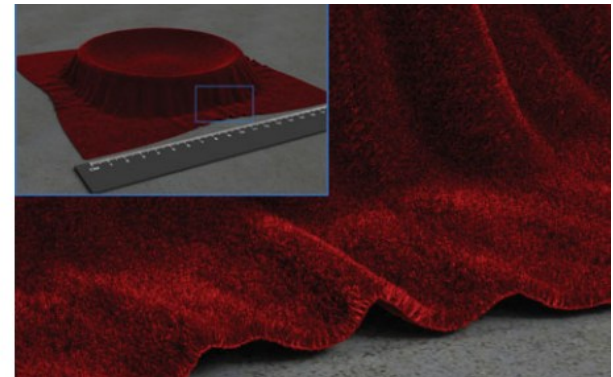
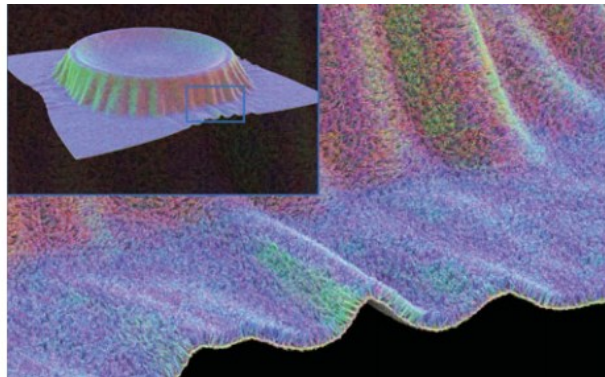
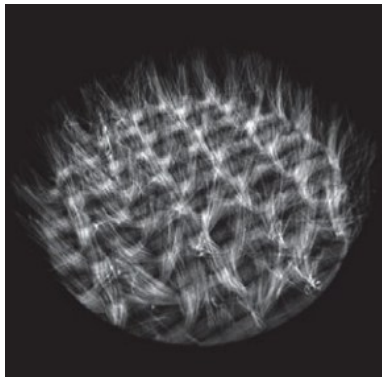


- Diffuse/specular reflectance components separation (polarization/color/illum. patterns) **[Shaffer 85]**, **[Nayar et al. JCV 97]**, **[Nayar et al. TOG 06]** $\Rightarrow$ diffuse component represents light refraction inside material structure $\Rightarrow$ fitting scattering models parameters to diffuse component.

Volumetric Models Acquisition

[Zhao et al. 11 TOG] – material geometry scanned by X-Ray Micro CT scanner (resolution 1024^2). Scattering information transferred to volumetric data by matching of several samples photographs.

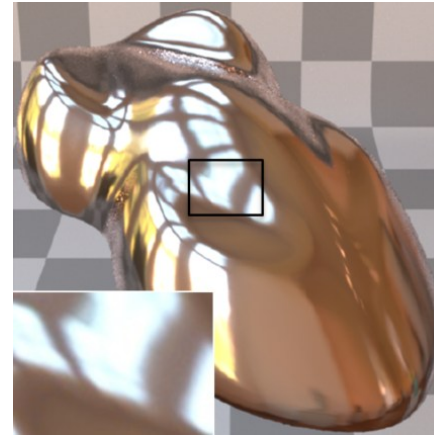
- ✓ realistic appearance
- ✗ assumes single material, lengthy rendering, limited dynamic range of the scanner



[Zhao et al. 11, ACM]

Practical Examples of BRDF Measurement

1. Highly specular materials



2. Extremely dark materials

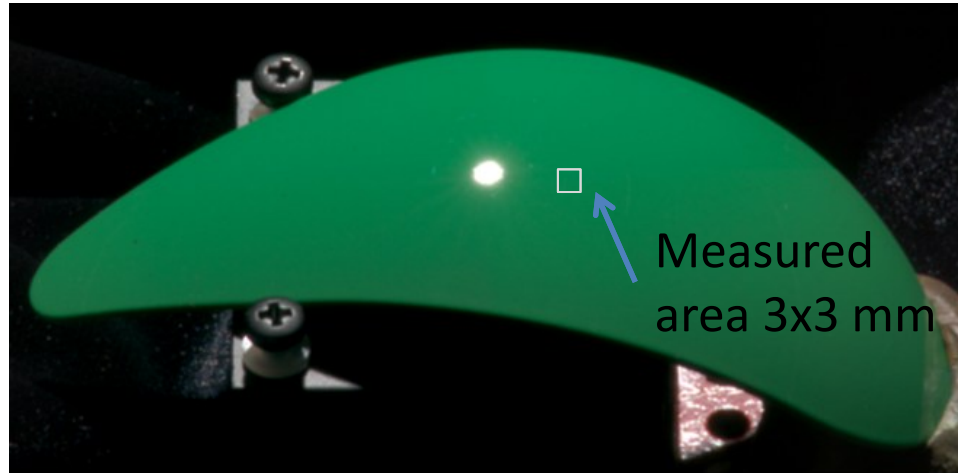


3. Translucency

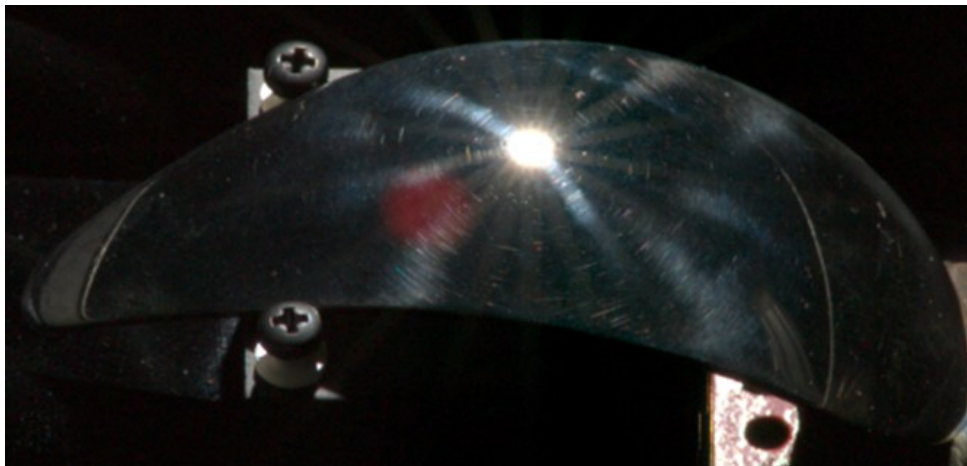


Specular BRDFs

Green paint with thick acrylic coating



Silver 925 (silver 92.5%, copper 7.5%)



Shape courtesy of Lechler S.p.a. Como, Italy



Coating industry needs replacement of health-risky chromium surfaces

- BRDF captured by averaging BTF images from UTIA goniometer
- Half-difference parameterization
- non-linear sampling along θ_h
- 108 images/samples

Specular BRDFs

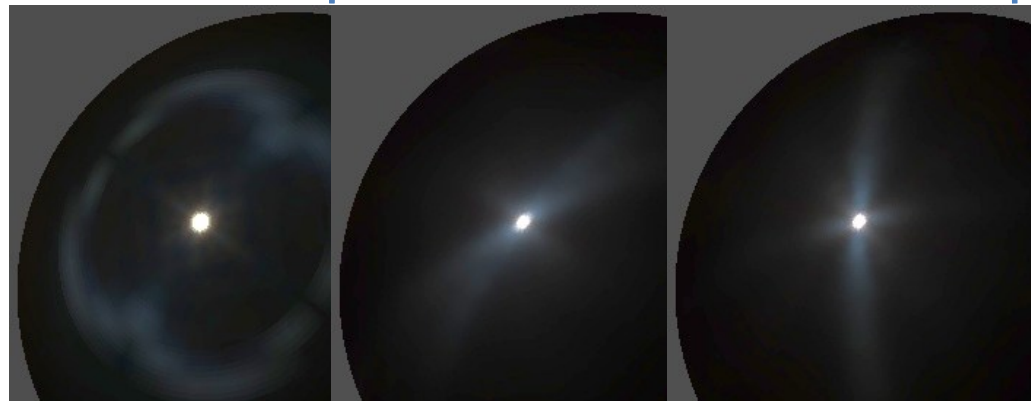
- anisotropy due to polishing
- light refraction on aperture
- lens flare artifact
- reflections of environment



Anisotropic artifacts

- due to insufficient polishing
- highlight perpendicular to scratches
- can be avoided by selecting appropriate

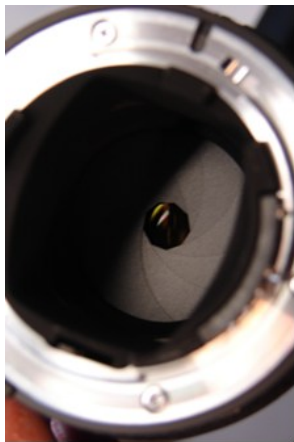
BRDFs from MERL database cannot be safely used as a reference



Specular BRDFs

Diffraction on aperture

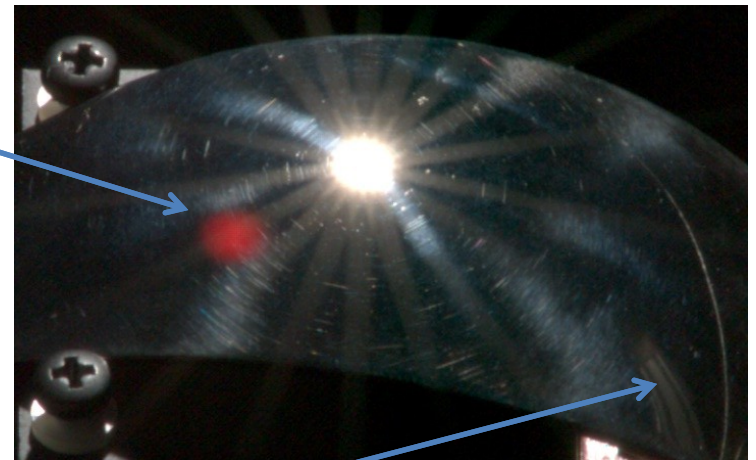
- wave effect of light $\Rightarrow$ light passing narrow slit
- bright streak perpendicular to aperture blades
- streaks in both directions $\Rightarrow$ odd number of blades $\Rightarrow$ twice so many streaks
- pronounced for narrow aperture and bright point-light
- solved by circular aperture



Specular BRDFs

Lens flare

- usually red spot due to inter-reflection in lens body
- the more elements the more pronounced
- Suppressed by a long lens hood



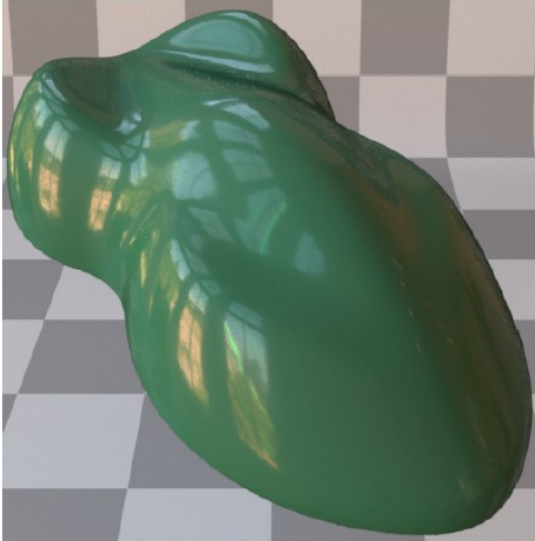
Reflection from environment

- mirror-like finishes reflect environment – diffuse black covering

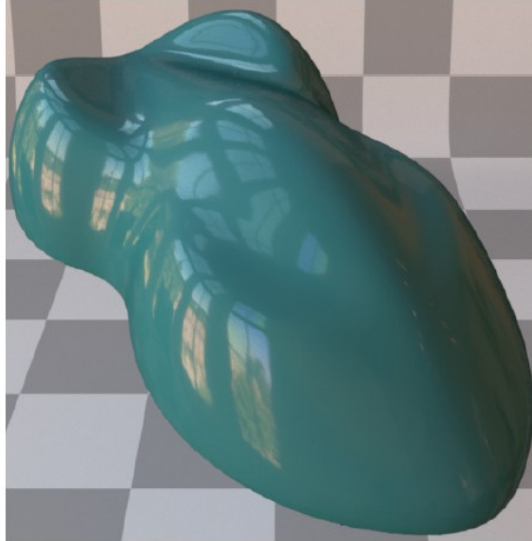
Specular BRDFs

[See Interactive Graphics Course - Cem Yuksel](#)

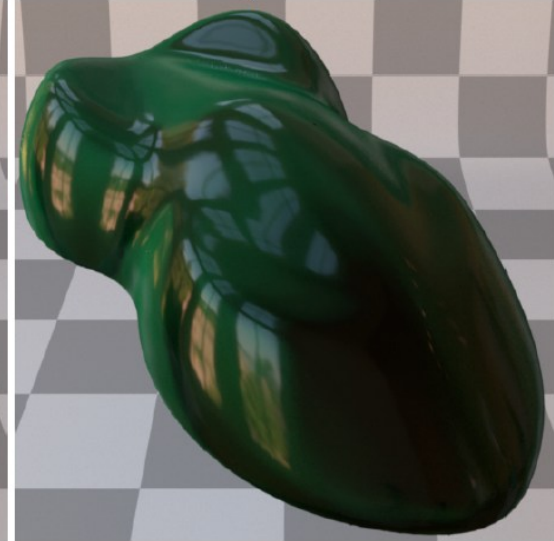
MERL *green-acrylic*



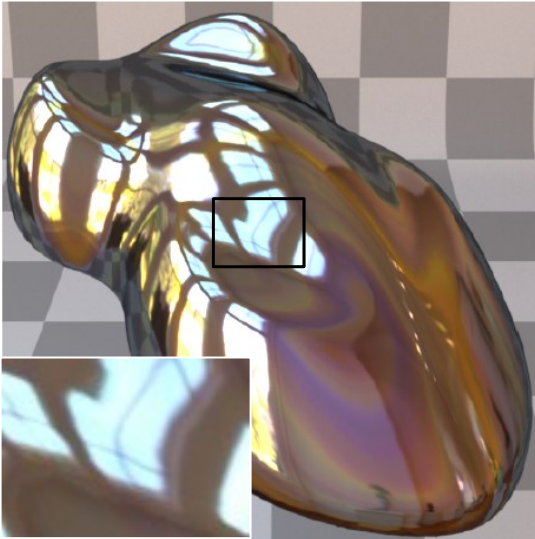
MERL *green-plastic*



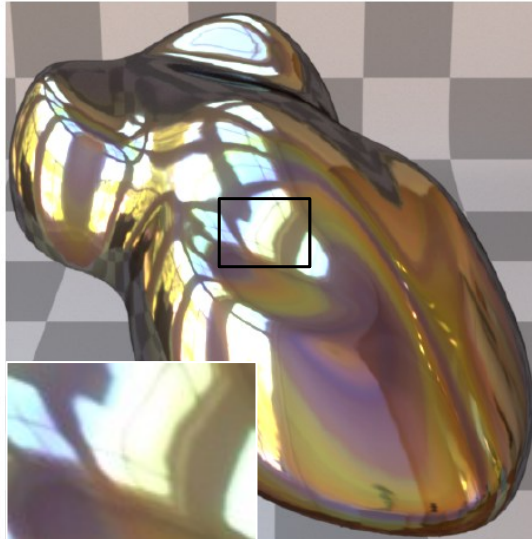
our measurement paint



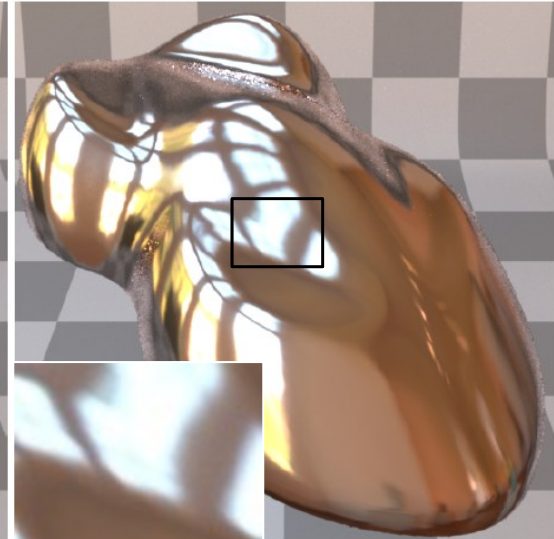
MERL *chrome*



MERL *steel*

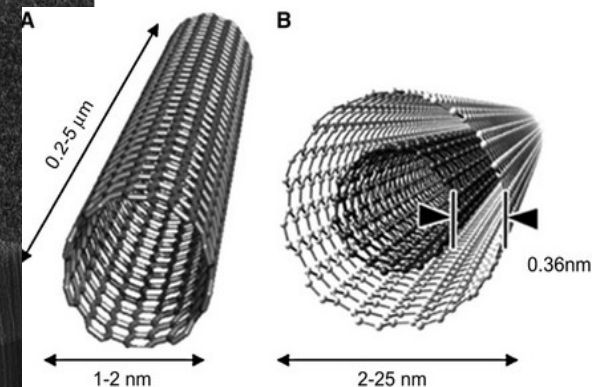
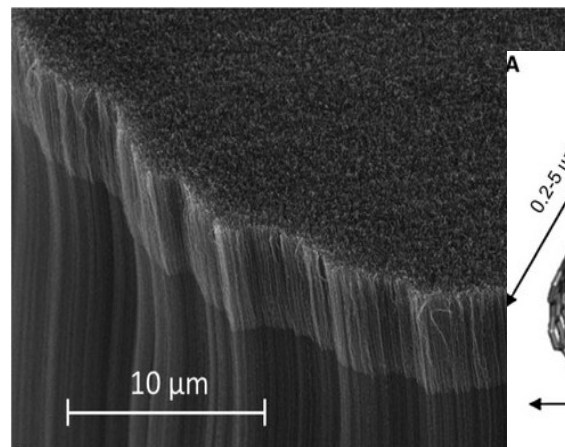
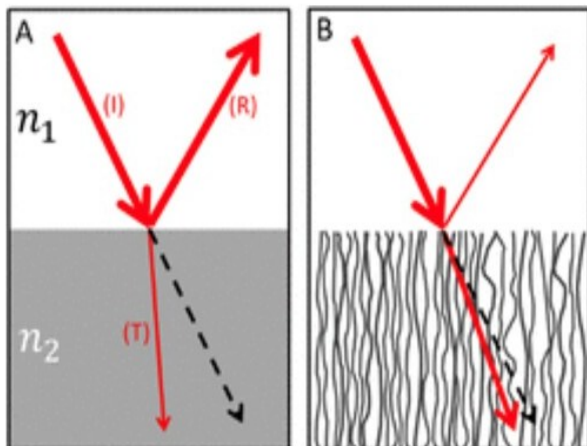
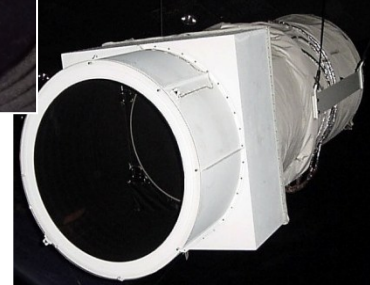


our measurement silver



Extremely Dark BRDFs

- Black materials used for stray-light suppression, calibration, imaging
- Appear similar under diffuse light
- Strong differences under directional illumination
- Ultra-blacks (VACNT) are costly and fragile
- Informed cost/performance analysis of alternative solutions



Extremely Dark BRDFs

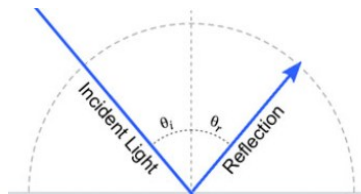
- **Vantablack (VACNT)** 40mm² **150€**

Paints:

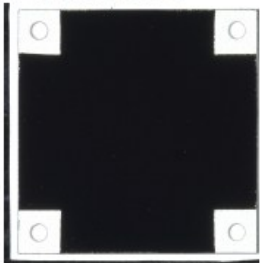
- **Musou Black paint** (brush-on) (100ml 40€) **~0,50€**
- **Acrylic paint** (spray) (500ml 5€) **~0,01€**
- **Chalkboard paint** (brush-on) (500ml 18€) **~0,01€**

Fabrics:

- **Black velvet** (1x1.45m 10€) **0,01€**
- **Musou fabric** (A4 20€) **0,52€**



Vantablack



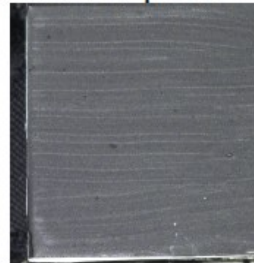
Musou paint



acryl paint



chalkb. paint



black velvet

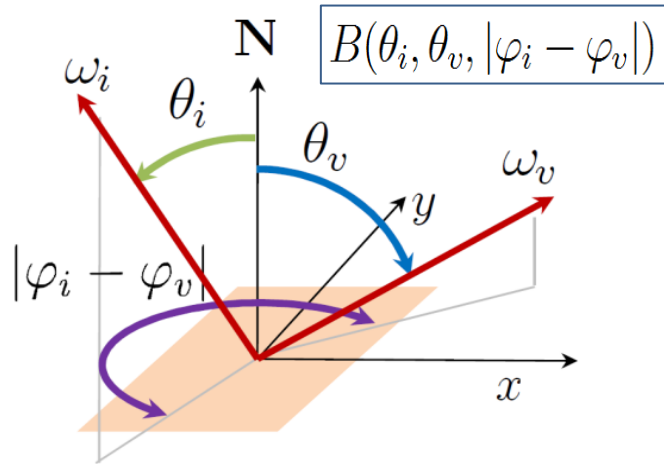


Musou fabric

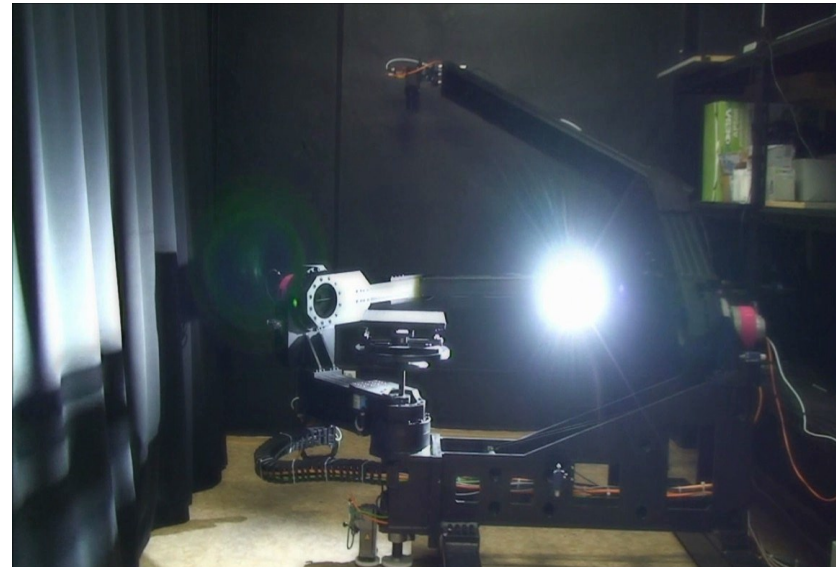
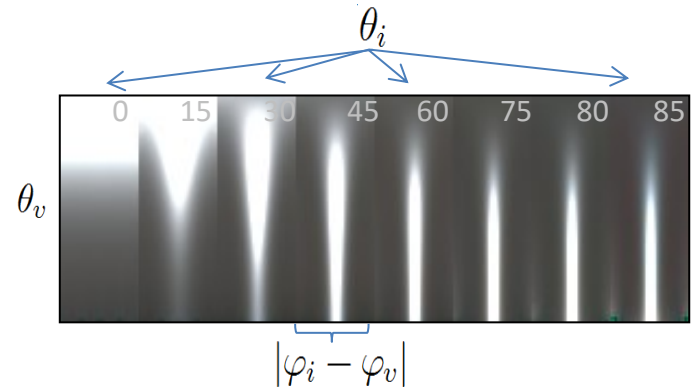


Extremely Dark BRDFs

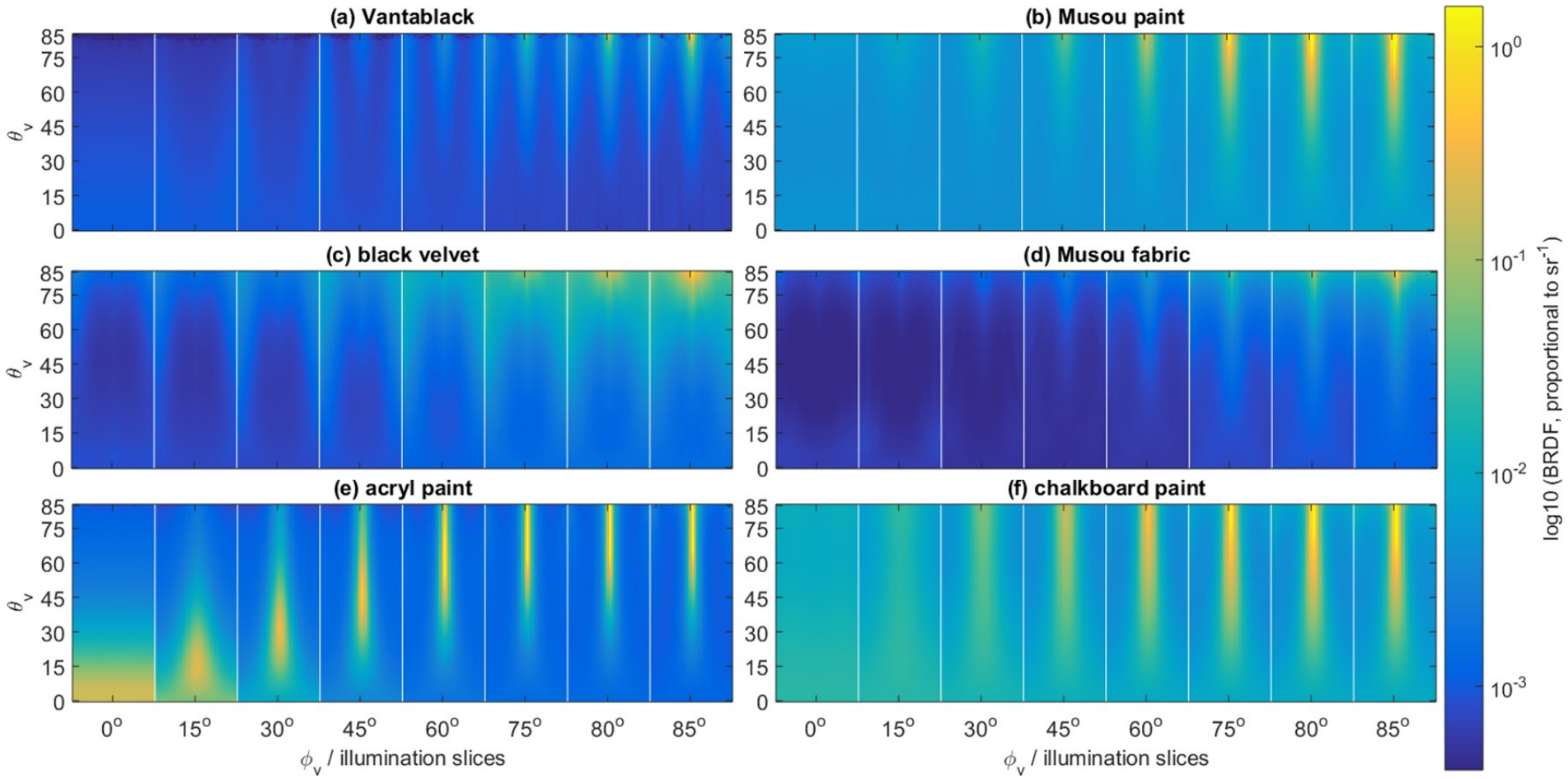
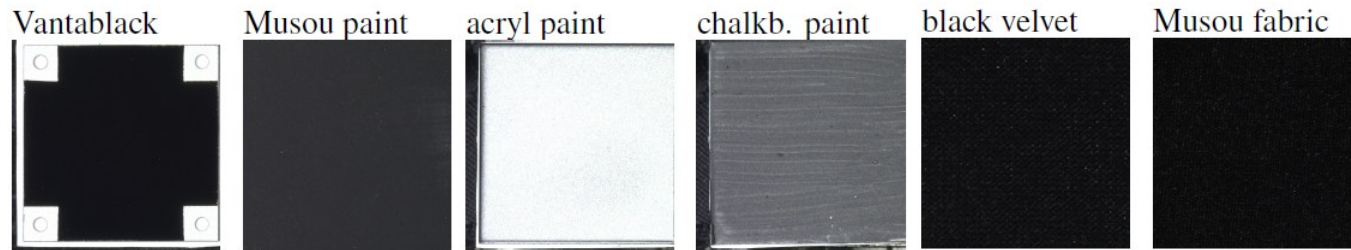
- Dense relative BRDF acquisition



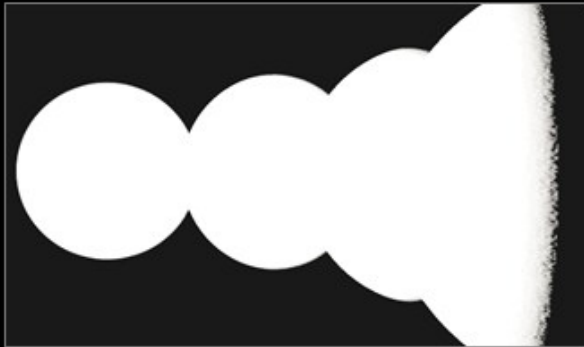
- UTIA high-precision goniometer
- Visible spectrum, RGB camera
- 24 488 samples per material
- Black materials – long exposure times – measurement time 96 hours
- Suppression and compensation of stray light



Extremely Dark BRDFs

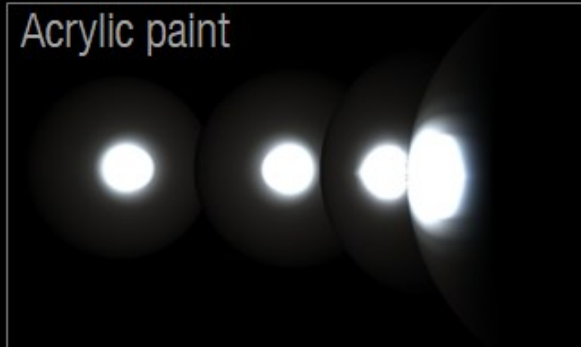


Extremely Dark BRDFs

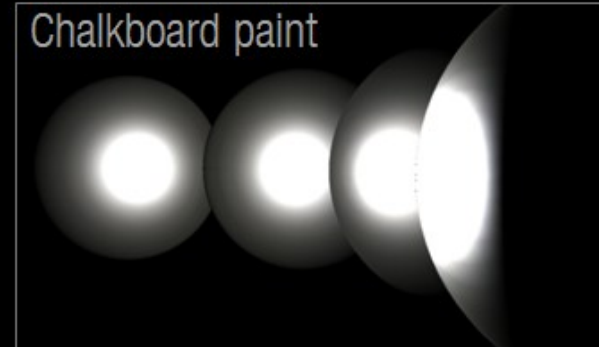


100x light intensity

Acrylic paint



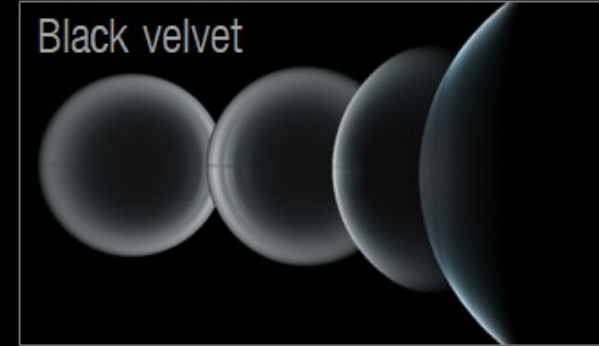
Chalkboard paint



Musou paint



Black velvet



Musou fabric



Vantablack



- Coatings reveal specular artifacts
- Fabrics and ultra-black remain visually dark

Extremely Dark BRDFs

- **Total hemispherical reflectance (THR)** - measures absolute darkness

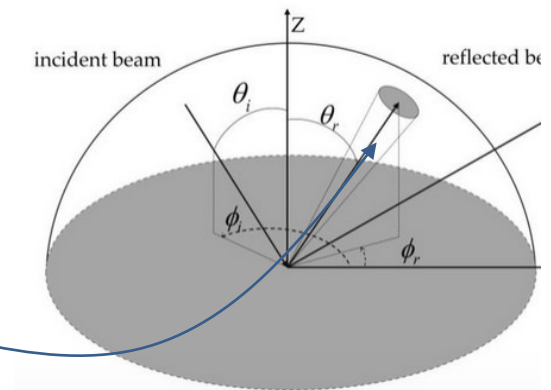
$$\rho(\omega_i) = \int_{\Omega_v} f_r(\omega_i, \omega_v) \cos \theta_v d\omega_v$$

- **Albedo**

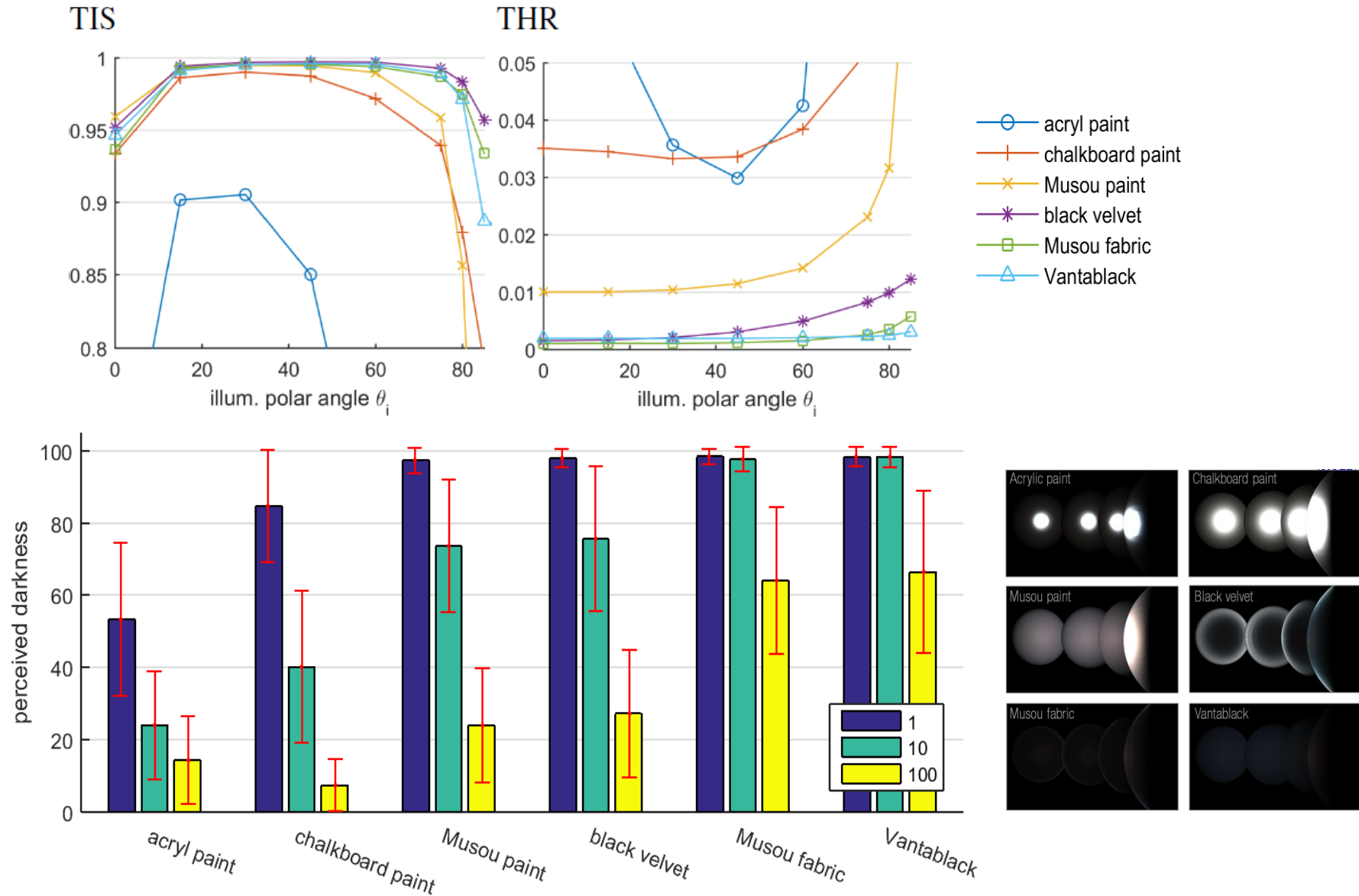
$$A = \int_{\Omega_i} \rho(\omega_i) \cos \theta_i d\omega_i$$

- **Total integrated scatter (TIS)** - measures scattering uniformity

$$\text{TIS} = \frac{R_d}{R_t} = \frac{R_d}{R_s + R_d}$$

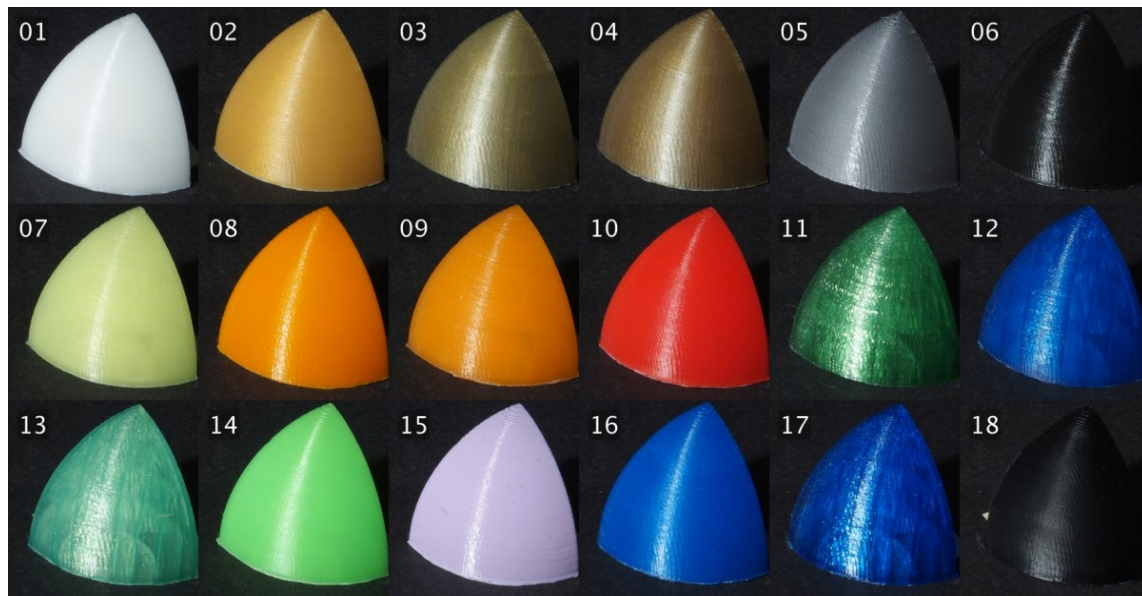


Extremely Dark BRDFs



Capturing Translucency of Printing Filaments

- 17 3D printing filaments of width 1.7 mm
 - 15 made from fifteen PLA (polylactic acid)
 - 2 made from PET (polyethylene terephthalate) (samples 13 and 17)



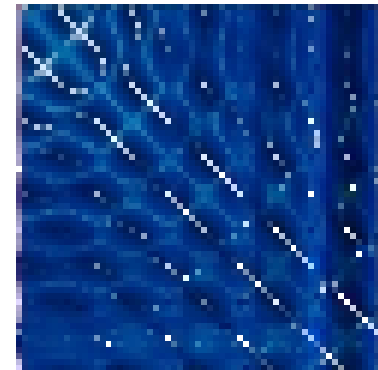
Capturing Translucency

- Sighttex Q material appearance capture device
- BRDF of 3660 values, 61 x 60 illumination/view directions
- BRDF of small spot on each filament for white and dark background



measurement spots

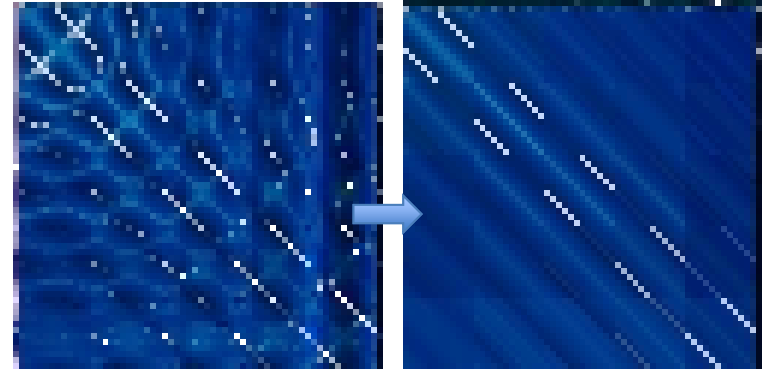
Viewing directions



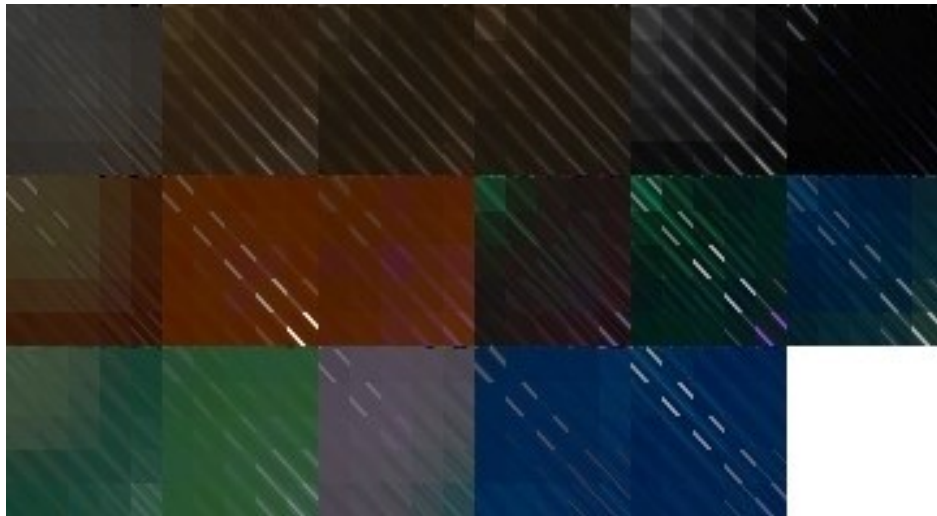
Illumination directions

Capturing Translucency

- Filtering raw BRDF data by enforcing light/view reciprocity and isotropy



BRDFs for white background



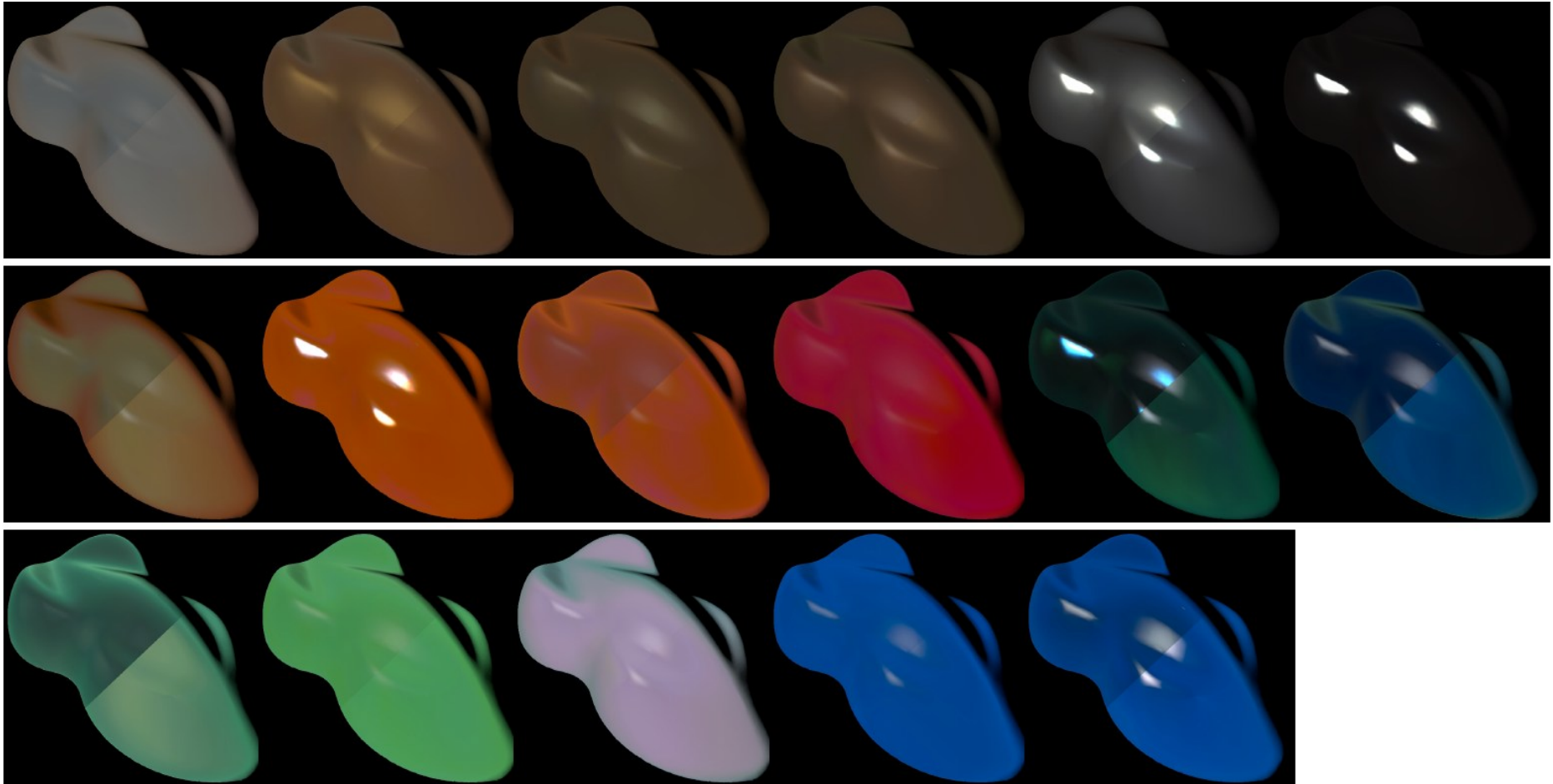
Translucency as white-dark difference



Capturing Translucency



BRDFs renderings for dark/white background



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